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UNIVERSITI TEKNOLOGI MALAYSIA

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DATABASE DESIGN DESCRIPTION: CONCEPTUAL DATABASE DESIGN

FOR Hasta Travel Vehicle Rental Management System

Group Name: DataATE

Members:

1. Angela Ngu Xin Yi A24CS0226
2. Gavenesh A/L Batumalai A24CS0076
3. Tan Xin Tian A24CS0198
4. Teoh Xin Yee A24CS0307
5. Toh Shee Thong A24CS0309

Stakeholder:

Encik Alif	23 January 2026
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1.0 Introduction

Hasta Travel and Tour SDN BHD is a vehicle rental company that serves exclusively to staff and students at Universiti Teknologi Malaysia (UTM). Facilitating seamless transportation for employees and students is the company's main goal. This project focuses on digitizing the company's current activities in an attempt to fit into modern digital norms. The project intends to simplify the leasing process by integrating a new digitalized system with their business model. This would reduce the administrative workload while also facilitating user access to services.

The company's current reliance on a manual booking method gives serious operational challenges. Under the current procedure, customers must complete a "WhatsForm" on the website and send individual WhatsApp inquiries. Time consuming availability checks and a slow response rate are just two of the inefficiencies caused by this manual method. Additionally, the absence of a centralized system raises the probability of human errors, such as misplaced communications or contradictory reservations, which combines both operational effectiveness and the general customer experience..

To address these challenges, this project proposes the development of a responsive web based vehicle rental management system. which will offer a real time dashboard displaying vehicle availability and dynamic pricing that adjusts based on seasonal demand. Secure data management will be handled through a MySQL database, ensuring the safe storage of user, staff, and vehicle information. Additionally, the system will automate communication by sending booking confirmations and invoices directly to the user's registered email address, eliminating the need for manual confirmation messages.

Firstly, the user will have a 24/7 real time vehicle availability update and an automated booking confirmation details sent through their emails. Next, the manager (system administration) will have a digital record keeping rather than needing to manually update their information and the workload of each staff will be reduced significantly. While established commercial platforms such as Penang Book, Wahdah and Socar offer fully digitalized vehicle rental services, the proposed system is distinct in its specific focus on the UTM ecosystem.

1.1 Definitions, Acronyms and Abbreviation

1.1.1 Definitions

Table 1.1.1.1 Definitions

Term	Definition
Admin	An authorized system user with elevated privileges to manage day-to-day operations including customer verification, booking approval, payment verification, and fleet management.
Manager	The highest-level system user with full access to all modules, including financial reports, user role management, and system configuration.
Blacklist	A status assigned to customers who have outstanding payments, penalties, or other violations, restricting them from making new bookings until issues are resolved.
Grace Period	A 30-minute buffer time after the scheduled return time before late penalties are applied.
Surcharge	Additional fees applied for pickups or returns at non-standard locations or during specific time periods.

1.1.2 Acronyms

Table 1.1.2.1 Acronyms

Acronym	Meaning
API	A set of protocols and tools that allow different software applications to communicate with each other.
CC	The engine displacement or volume of an engine's cylinders, measured in cubic centimeters.
IC	Malaysian national identity card (MyKad) required for customer verification.
JPJ	The Road Transport Department of Malaysia, responsible for vehicle registration and licensing.
UTM	Universiti Teknologi Malaysia

1.1.3 Abbreviation

Table 1.1.3.1 Abbreviation

Abbreviation	Meaning
IC No	Identity Card Number (00000-00-0000)
Matric No	Matric Number (X00XX0000)

2.0 DFD (to-be)

2.1 Context Diagram

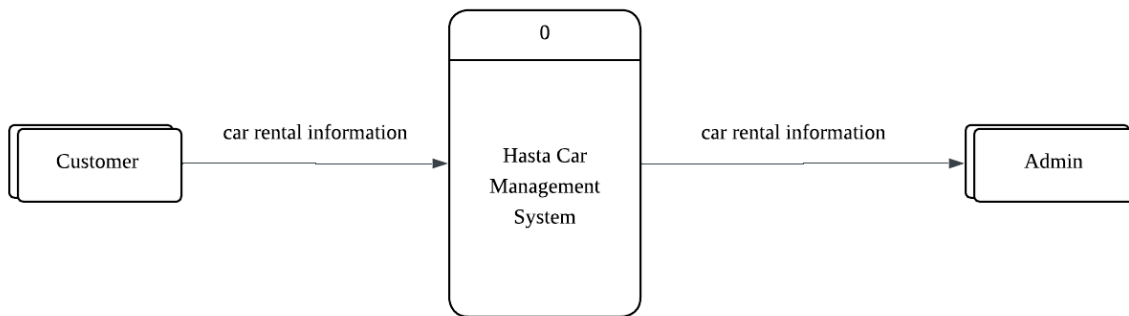


Figure 2.1: Context Diagram

2.2 Diagram 0

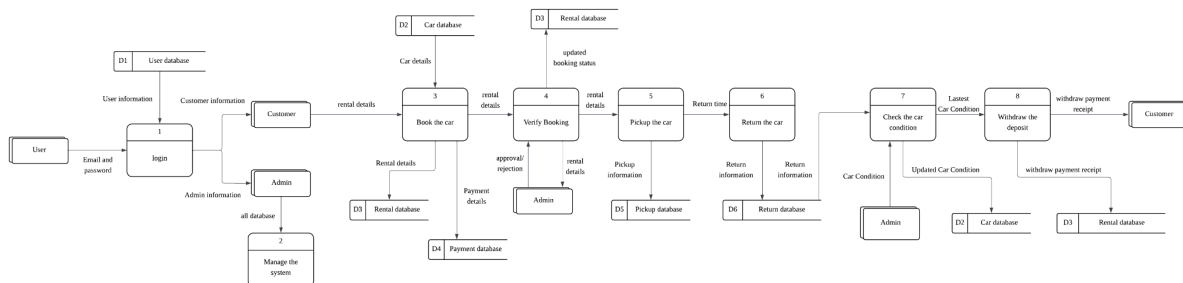


Figure 2.2: Diagram 0

2.3 Child Diagram

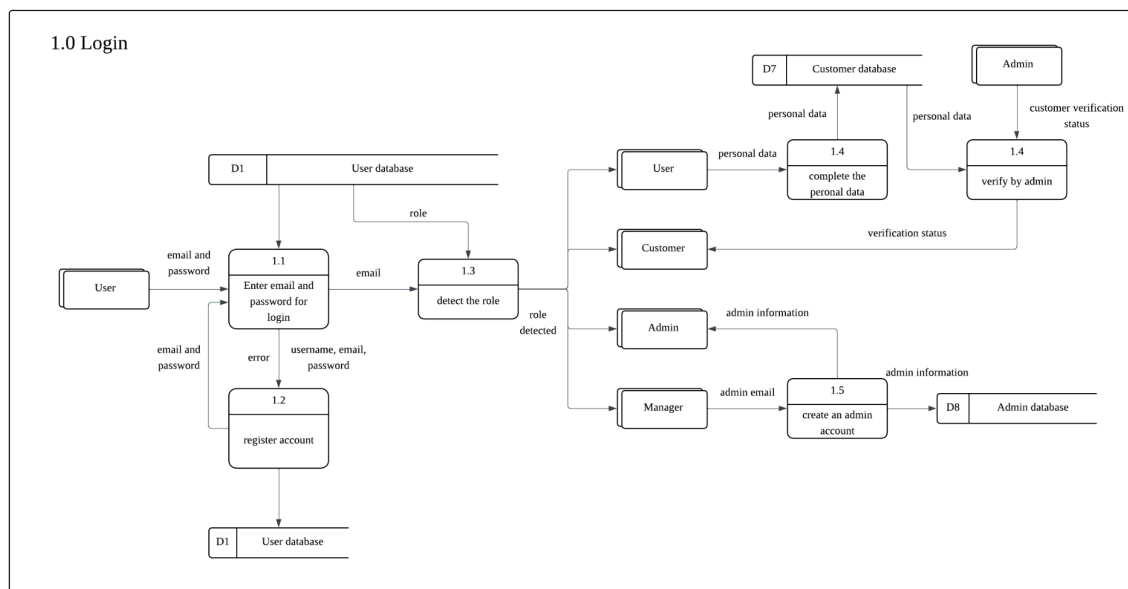


Figure 2.3.1 Child Diagram for Process 1: Login

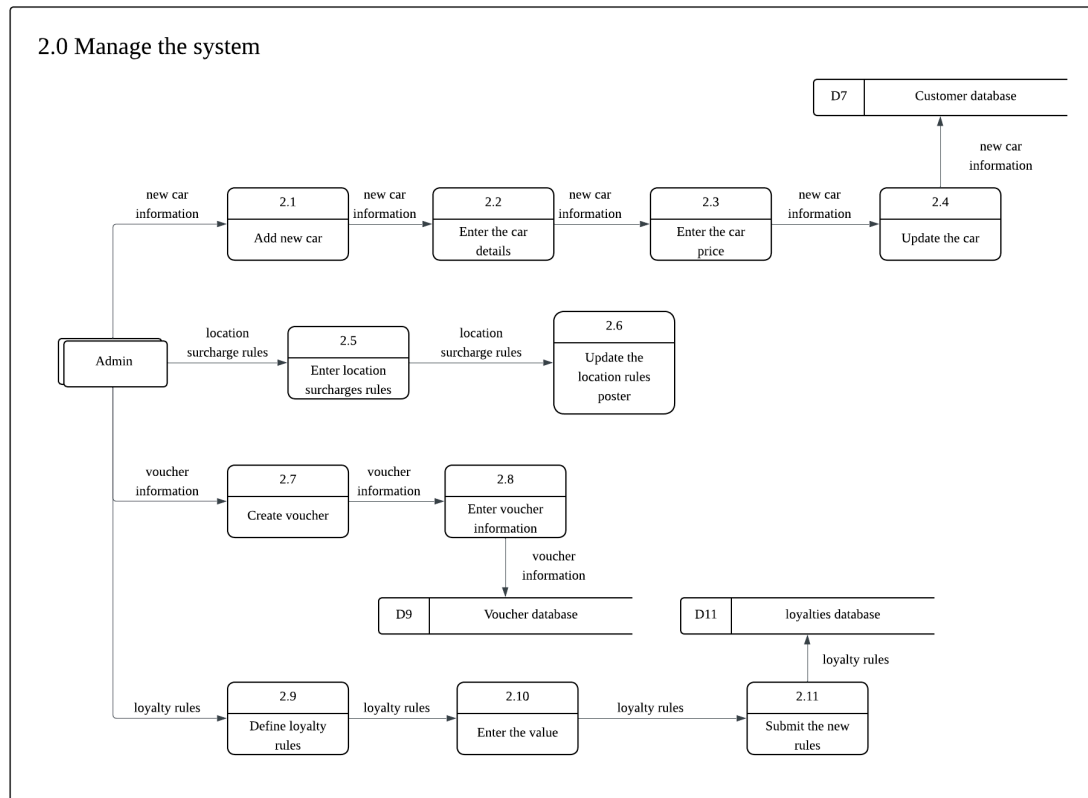


Figure 2.3.2 Child Diagram for Process 2: Manage the System

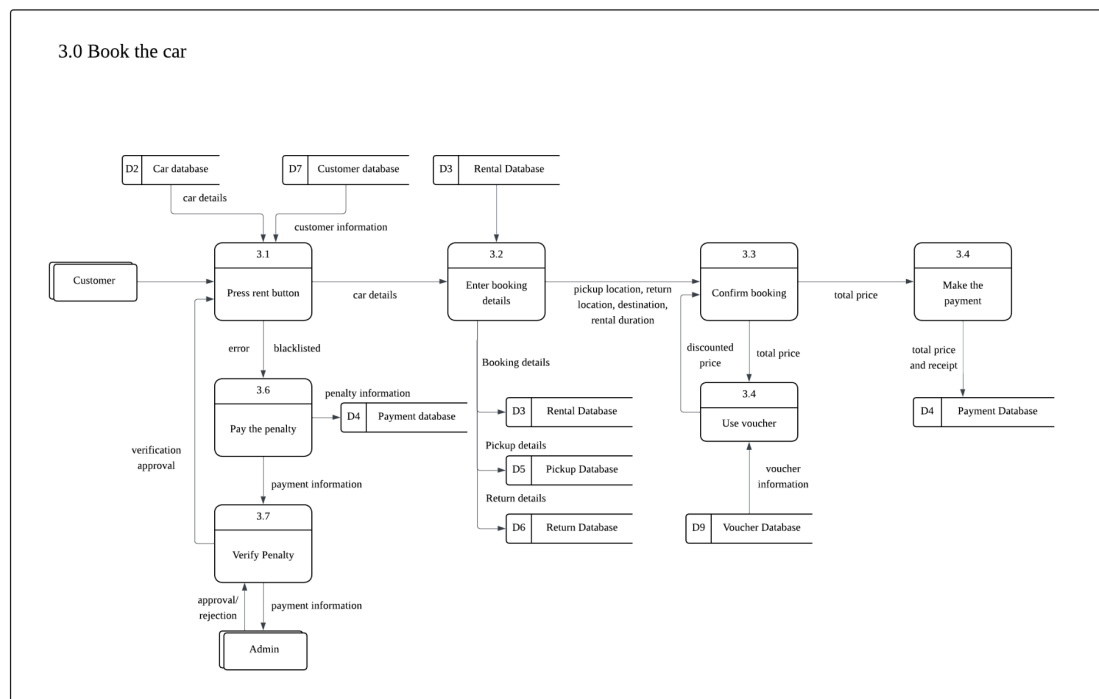


Figure 2.3.3 Child Diagram for Process 3: Book the Car

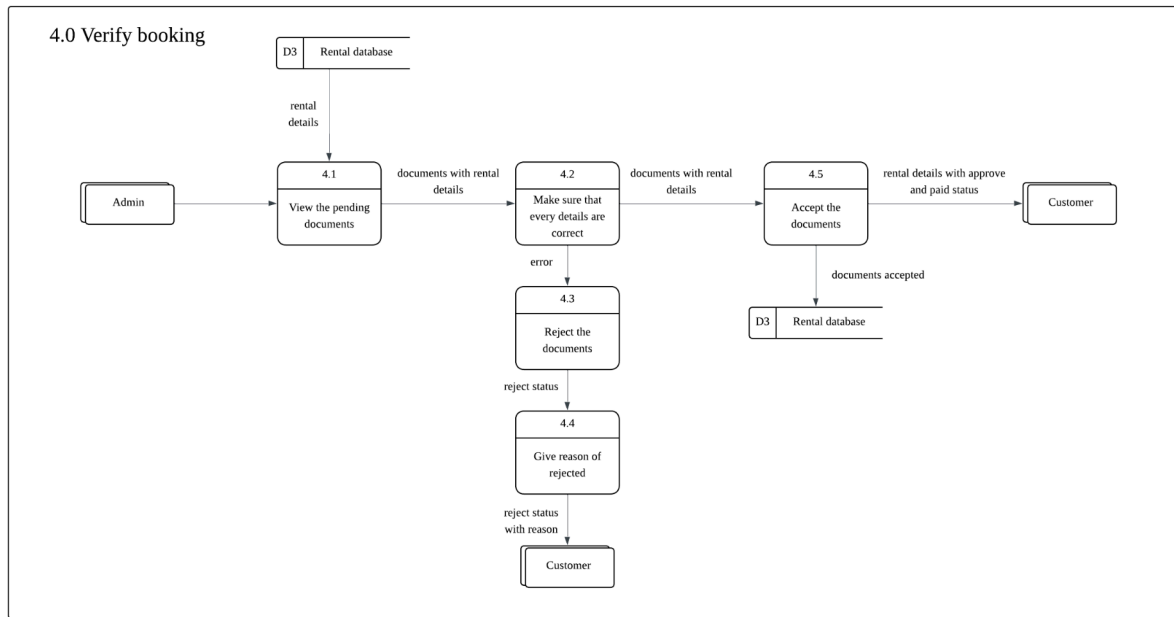


Figure 2.3.4 Child Diagram for Process 4: Verify Booking

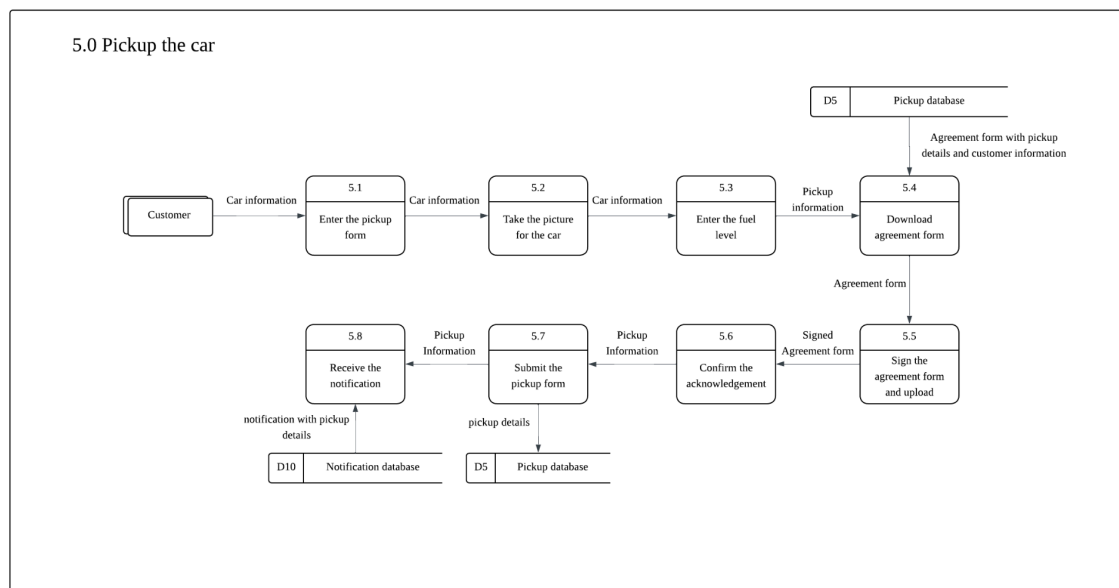


Figure 2.3.5 Child Diagram for Process 5: Pickup the Car

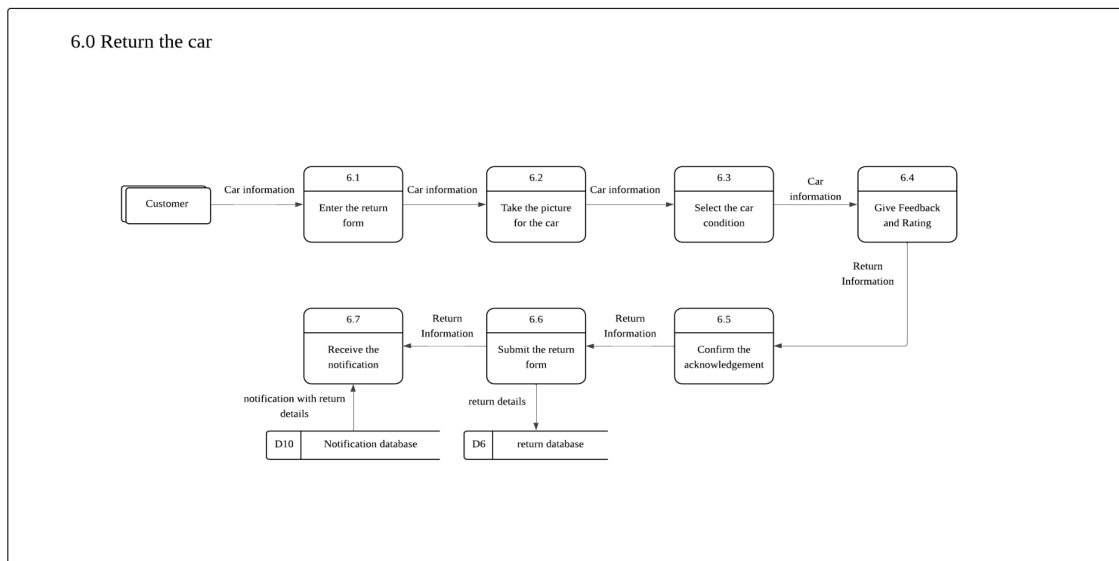


Figure 2.3.6 Child Diagram for Process 6: Return the Car

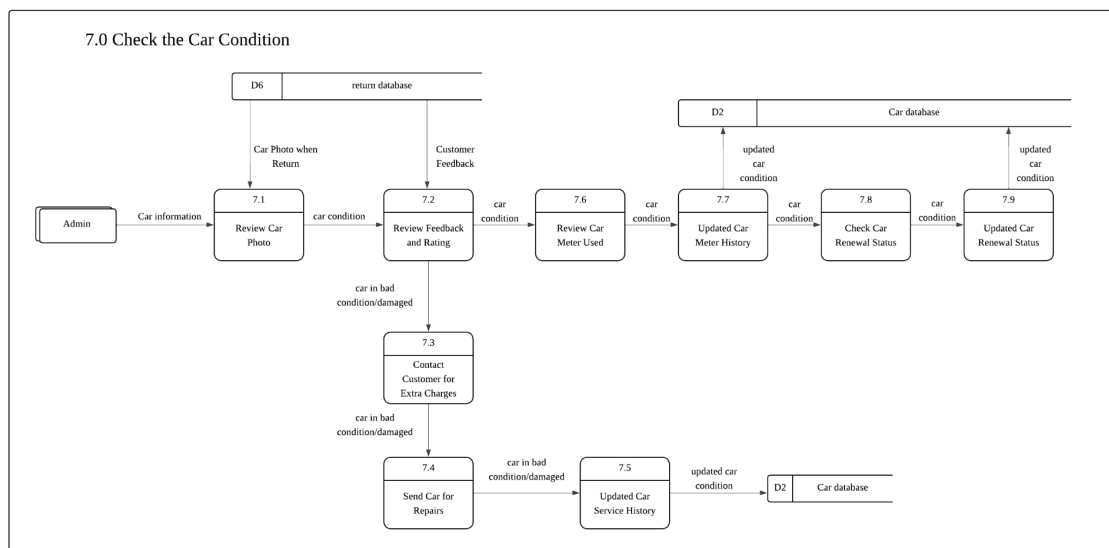


Figure 2.3.7 Child Diagram for Process 7: Check the Car Condition

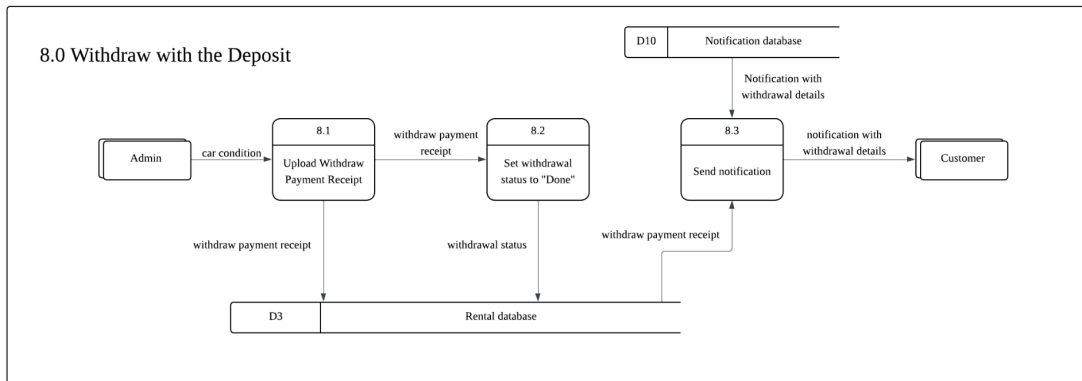


Figure 2.3.8 Child Diagram for Process 8: Withdraw the Deposit

3.0 Data and Transactional Requirements

3.1 Proposed business rule

This section outlines both the business rules and also the data and transactional requirements that the new Hasta Travel and Tours System need to attain.

3.1.1 User Management Module

1. All new public users are registered with the customer role by default
2. Customers must complete their specific profile data which are driving license, Identification Card (IC) or passport and matric card before they are permitted to book a vehicle
3. Uploaded documents require manual approval by an admin or manager
4. Admins or managers can approve or reject these documents and a reason is needed when rejection occurs
5. When the customers change the IC number or matric no, they have to re-upload the related documents
6. Booking privileges are blocked until documents are approved
7. Customers with outstanding penalties are marked as blacklisted and cannot make new bookings until they upload a payment receipt which is then verified by an admin

3.1.2 Fleet Management Module

1. The system designates vehicles in two primary states which are 'active' for those currently in the operational fleet and 'retired' for those permanently inactive or the owner does not want to rent the vehicle anymore
2. Vehicle availability is tracked dynamically which 'available' indicates readiness for booking while 'unavailable' indicates the vehicle is currently rented or disabled
3. A maintenance status overrides standard availability which prevents bookings when a vehicle is undergoing repairs
4. Rental prices are defined individually for each vehicle across standard durations which are one hour, three hours, five hours, twelve hours and one day. The system also applies different pricing tiers for weekends versus weekdays to accommodate demand fluctuations
5. The system records detailed owner information for each vehicle
6. Critical documents such as the vehicle grant, road tax and insurance cover note are uploaded and tracked digitally
7. The system actively tracks expiry dates for road tax, insurance and other recurring obligations. Admin can set due date and receive alerts for upcoming renewals
8. Service records log details of maintenance work, vendor name, cost and date while meter updates records tracks the odometer readings periodically to monitor vehicle usage

3.1.3 Booking Management Module

1. Customers must have a complete profile with all approved uploaded documents before they can initiate a booking

2. The system enforces a conflict check to strictly prevent any booking that might overlap with an existing pickup or return for the selected vehicle
3. The system applies additional fees for pickups or returns that occur at designated surcharge locations such as outside campus or during specific surcharge times. Surcharge rules are configurable via the admin and manager panel
4. Customers are only permitted to cancel a confirmed booking if the scheduled pickup time is more than 24 hours away.
5. Cancellations are eligible for a refund which is processed manually within 3 to 5 business days
6. If a booking using a voucher is cancelled, the voucher will be automatically restored for future use
7. Both pickup and return processes require a digital handover where the customer must upload four angle photos of the vehicle, a dashboard photo and note any damage
8. Fuel levels are recorded at both ends of the transaction to calculate penalties
9. Customers must digitally acknowledge the rental agreement at pickup
10. Customers are required to fill the ratings and feedback at return
11. Late return is calculated only after a 30-minute grace period from the scheduled return time and RM50 would be charged per hour
12. Fuel shortage penalties are applied based on the number of fuel bars missing as recorded during inspection which are RM10 per missing fuel bar
13. The penalties are subtracted from the deposit. If the penalty is less than deposit, the remaining deposit is refundable

3.1.4 Payment Management Module

1. A deposit is mandatory and collected for every single booking iteration
2. Admin must visually verify the uploaded receipt image before a booking can be confirmed
3. Rejected receipts will trigger a re-upload request or booking cancellation. Only “incorrect amount” triggers a refund workflow while other reasons typically require re-upload
4. The system records the specific admin ID who verified each payment for accountability

3.1.5 Loyalty and Rewards Module

1. Customers earn 1 loyalty stamp for every completed rental transaction with a duration of 9 hours or more
2. Customers can redeem collected stamps for vouchers based on defined tiers. Redemption immediately deducts the stamp count from the customer’s balance
3. Vouchers can optionally grant a percentage discount or specific free hours
4. A voucher can only be used once
5. If the booking is cancelled or rejected, the voucher will be automatically restored for future use
6. Admin and manager can assign voucher to the customers

3.1.6 Report and Analytics Module

1. Admin dashboard focuses on operational or daily tasks while manager dashboard focuses on strategic performance
2. Revenue is aggregated into Daily (Revenue vs Expenses) and Monthly (Total Sales Count and Amount) views
3. The dashboard has fleet status overview which shows whether the fleet are available, rented or under maintenance
4. Feedback Scores view the vehicle rating especially those ratings below 3 stars to alert admin and manager

3.2 Proposed Data & Transactional Requirements

3.2.1 User Management Module

User Module serves as the centralized identity management component of the Hasta Travel & Tours Rental Management System. It securely manages all registered entities including Customers, Admin and Manager. Moreover, it controls authentication, authorization and profile information.

3.2.1.1 Data Requirements

Users

Hasta Travel Vehicle Rental Management System maintains user accounts for all individuals accessing the platform. Each user is uniquely identified by ID and must register with a valid and unique email address with password. Stored data includes, ID, email, username, password, role and account status.

The user role attribute determines the level of system access and dashboard features available after login. Roles include customer, admin and manager. User roles are not fixed at registration and may be updated by authorized personnel. A Manager has the authority to update a user's role directly within the User table, for instance, upgrading a Customer to a Manager. This design enables flexible role management without requiring the creation of new user accounts.

Customers

Customer records represent Universiti Teknologi Malaysia (UTM) students or staff using the booking service. Each customer is uniquely identified by a customer ID. Stored data includes customer ID, full name, nationality, matric or staff card image, matric or staff number, identity card or passport image, identity card or passport number, gender, faculty, residential college, address, phone number, driving license image, driving license expiry date, bank name, bank account, emergency contact information including name, relationship with customer and phone number, document verification status and customer status. Customer data is linked to the corresponding user account.

Admin

Admin members are system users with additional privileges and created through Manager actions. Each admin is uniquely identified by an admin ID. Stored data includes admin ID, phone number, position and admin status. Admin accounts are linked to user credentials for authentication.

3.2.1.2 Transaction Requirements

Data Entry

1. Register a new user account with default role Customer
2. Promote an existing user to Manager by updating the role attribute
3. Create Admin account through Manager authorization

Data Update/Deletion

1. Update a user's role, for example, Customer to Manager
2. Activate or deactivate Admin privileges
3. Update user credentials and profile information
4. Update document verification status from pending to approved or rejected
5. Deactivate user accounts without deleting historical records

Data Queries

1. Retrieve user role during login to determine access rights
2. Load role-specific dashboard and permissions
3. Check email uniqueness during user registration
4. Handle password reset requests using registered email
5. Verify Manager authorization before allowing admin creation
6. List users based on role for manager management

3.2.1.3 Local Conceptual Data Model (ERD)

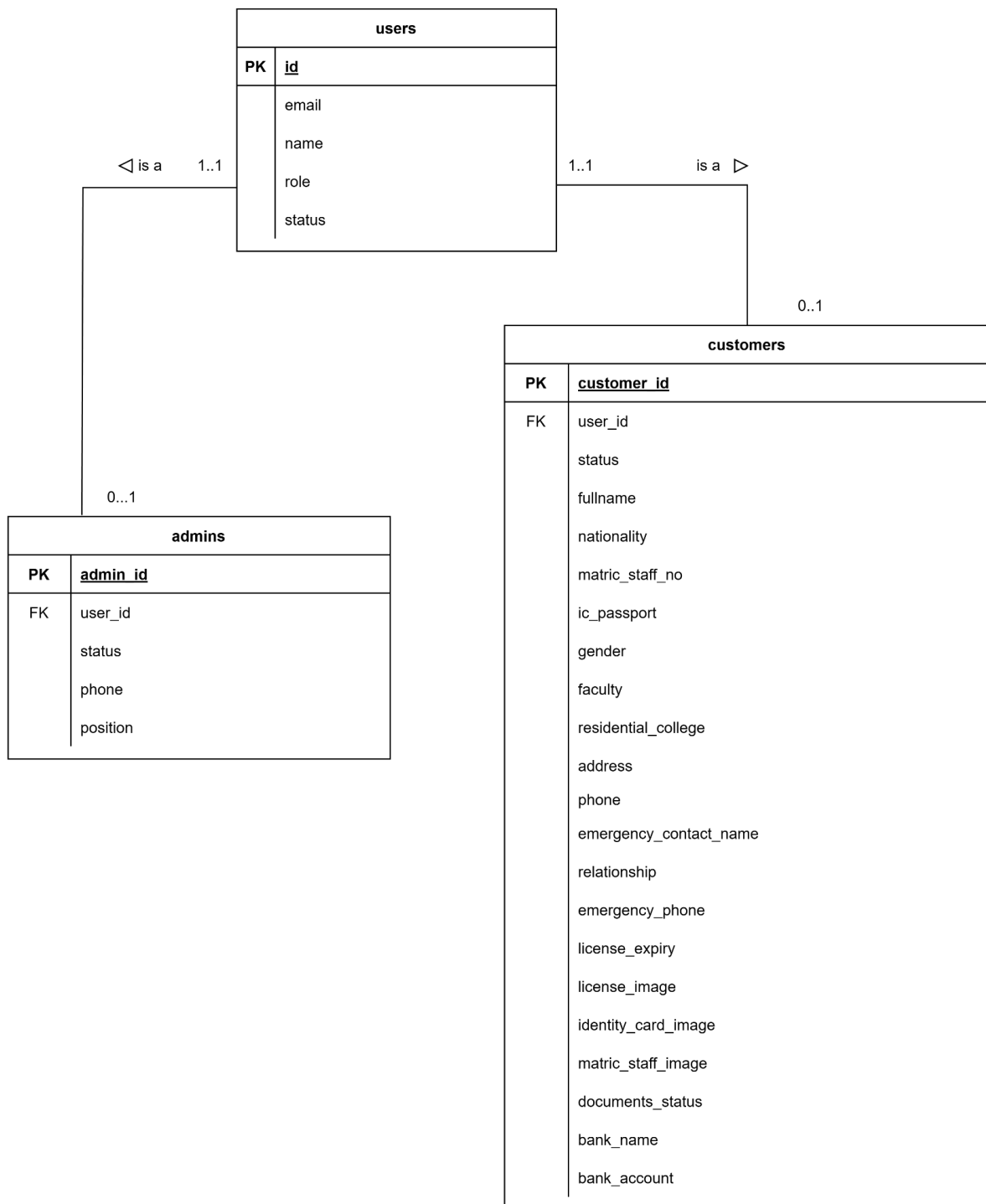


Figure 3.2.1.3.1 Local Conceptual Data Model for User Management Module

3.2.2 Fleet Management Module

This section outlines the vehicle module transaction requirements, which manage vehicle information, availability and status in the Hasta Travel & Tours Vehicle Rental Management System.

3.2.2.1 Data Requirements

Vehicle

Each vehicle is uniquely identified by vehicle plate number. Stored data includes general information namely, plate number, model, colour, seat, vehicle mileage, production year, engine, engine CC, fuel type, transmission type, drive type, maximum speed, torque and power, pricing rules, manufacturer, operating state and owner details such as name, identity card number, age, gender, race, citizenship, grant image path, roadtax image path, cover note image path and their respective expiry dates.

Operational data includes status, availability status and maintenance status. Status defines whether the vehicle is a valid member of the fleet that can be rented or not while availability status and maintenance status are used to control booking and fleet management activities.

Vehicle Photos

The system stores multiple photos for each vehicle to provide visual reference for system users. Each photo record is uniquely identified by a photo ID and is linked to a specific vehicle plate number. Stored data includes photo ID, vehicle plate number and label. This separation allows multiple images to be associated with a single vehicle without duplicating vehicle data.

Renewal Reminder

The renewal reminder records are used to manage expiry-related vehicle documents such as road tax and insurance. Each renewal reminder is uniquely identified by a reminder ID and is linked to a specific vehicle. The data stored for each renewal reminder includes reminder ID, service type (such as insurance or road tax), due date, reminder notes and reminder status. This separation allows a vehicle to have multiple renewal reminders and enables the system to track upcoming expiries without affecting core vehicle data.

Service History

The system stores maintenance and meter update information for vehicles in the system. Each service history record is uniquely identified by a service ID and is linked to a specific vehicle. Stored data includes service ID, vehicle name, completed date, current odometer reading, previous odometer reading, previous odometer update date, service type, service vendor, service description and service amount. This supports multiple service and meter update records for the same vehicle, allowing the system to maintain a complete history of vehicle maintenance and mileage updates.

3.2.2.2 Transaction Requirements

Data Entry

1. Register a new vehicle in the admin or manager site of the system with complete technical, legal and operational details
2. Upload one or more photos for a vehicle
3. Define hourly rental prices for a vehicle based on different time durations
4. Insert a renewal reminder record and record the service type, due date and optional notes
5. Insert a new service history record when a vehicle undergoes maintenance
6. Insert a new meter history record when the vehicle odometer is updated such as after rental return or inspection

Data Update/Deletion

1. Update vehicle details such as availability status, mileage or maintenance status
2. Update or replace vehicle photos
3. Modify the pricing rate
4. Deactivate a vehicle from the system without removing historical booking data
5. Update the renewal reminder status, due date and note when the expiry date is approaching or when a reminder has been processed
6. Delete a renewal reminder record if it is entered incorrectly or no longer required
7. Update service history details if corrections are required
8. Update previous odometer information automatically when a new meter update record is added
9. Delete a service history record if it was entered incorrectly without affecting other service records

Data Queries

1. Retrieve a list of available vehicles based on date and time
2. Retrieve vehicle details and associated photos for display
3. Retrieve applicable rental duration pricing for a selected vehicle
4. Check vehicle availability before confirming a booking
5. List vehicles by status (available, unavailable, under maintenance)
6. Retrieve vehicle usage data for reporting and analysis
7. Retrieve renewal reminders that are approaching the due date
8. Retrieve renewal reminders based by service type and based on vehicle plate and reminder status
9. Retrieve the complete service history of a vehicle based on vehicle plate number
10. Retrieve meter update records to track odometer changes over time

3.2.2.3 Local Conceptual Data Model (ERD)

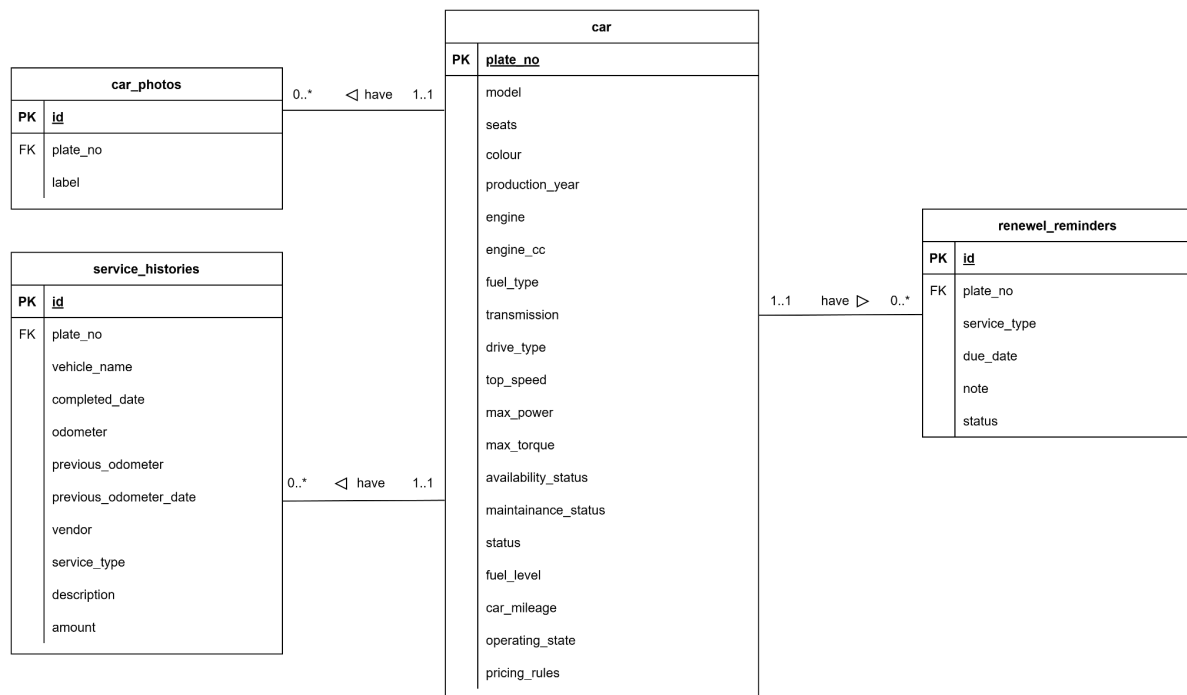


Figure 3.2.2.3.1 Local Conceptual Data Model for Fleet Management Module

3.2.3 Booking Management Module

The Booking Module manages the complete vehicle rental lifecycle, from reservation creation to vehicle pick up and return. It coordinates booking details, rental periods, vehicle assignment and booking status updates. This module ensures that vehicles are allocated correctly, rental rules are enforced and booking records are maintained for operational and reporting purposes. It allows customers to create, manage and track vehicle bookings while allowing admin and manager to verify, approve and monitor booking activities.

3.2.3.1 Data Requirements

Booking

Each rental is uniquely identified by a rental ID. Stored booking data includes rental ID, booking status, withdrawal receipt, withdrawal status, destination, invoice, payment status, fuel penalty and late penalty. Booking status indicates the current stage of the rental process such as pending, verified, renting, done and rejected. Each rental has one payment and one voucher. Each booking is associated with one vehicle and a customer can make multiple bookings.

Pick Up

Pick up records capture information related to vehicle handover to the customer. Each pick up record is uniquely identified by a pick up ID and linked to a rental. Stored data includes pick up ID, rental start date and time, pick up location, vehicle condition images at pick up, dashboard image, damage image if any damage exists on vehicle before pick up, description

about damage part, fuel level at pick up, agreement form, acknowledgement status and pick up report. This information ensures proper documentation of the vehicle condition before the rental begins.

Return

Return records store information related to vehicle return at the end of rental period. Each return record is uniquely identified by a return ID and linked to a rental. Stored data includes return ID, rental end date and time, return location, vehicle condition images at return, dashboard image, damage image if any damage exists in the rental period, damage remarks, fuel level at return, service rating, vehicle condition rating, feedback, acknowledgement of the images for different views and return report.

3.2.3.2 Transaction Requirements

Data Entry

1. Create a new booking by selecting a vehicle, rental period and destination
2. Record vehicle pickup details during handover
3. Record vehicle return details upon rental completion

Data Update/Deletion

1. Update booking status throughout the rental lifecycle
2. Update late penalty if the customer returns the vehicle late
3. Update fuel penalty when the return fuel level is less than pickup fuel level
4. Cancel or close bookings without deleting historical records

Data Queries

1. Retrieve booking details for a specific customer or vehicle
2. Check vehicle availability for a selected rental period
3. Retrieve pickup and return records linked to a booking
4. List verified, in progress, completed or cancelled bookings
5. Retrieve booking history for customer dashboards

3.2.3.3 Local Conceptual Data Model (ERD)

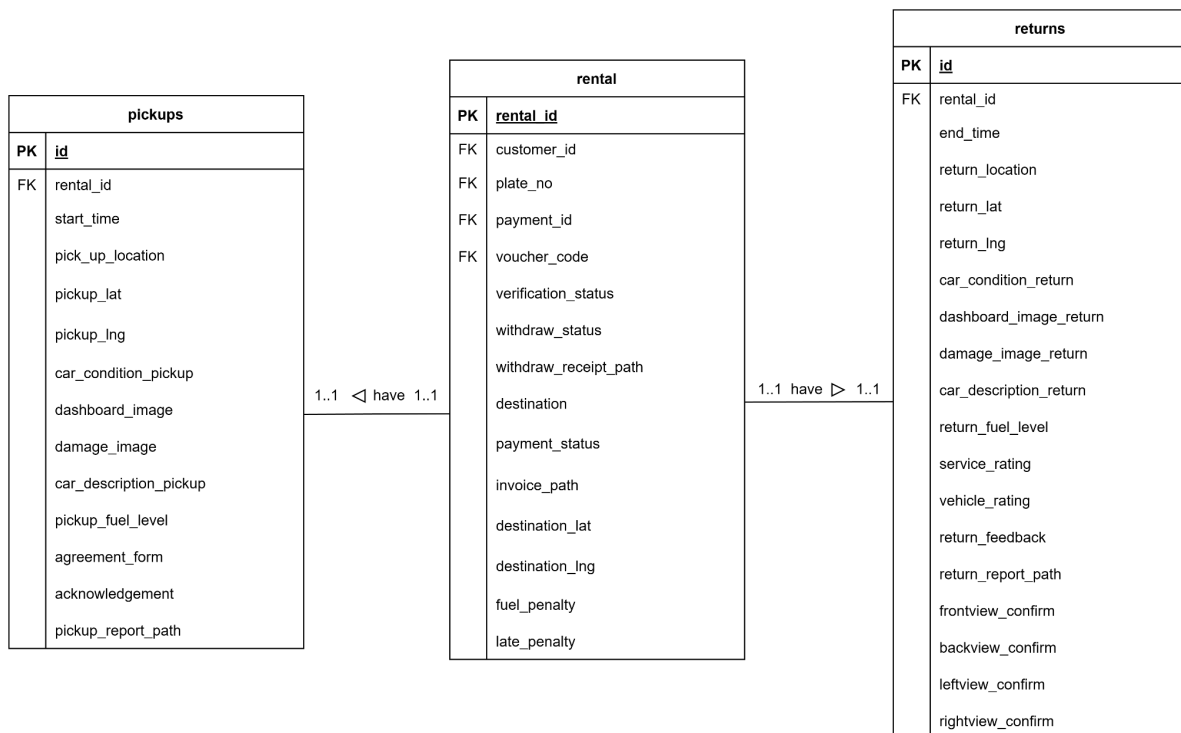


Figure 3.2.3.3.1 Local Conceptual Data Model for Booking Management Module

3.2.4 Payment Management Module

This module manages all financial transactions within the Hasta Travel & Tours Vehicle Rental Management System. It handles rental payments, deposits, penalties and refunds while supporting manual payment verification by administrators or managers. It ensures that all payments are properly recorded, verified and linked to the relevant booking and customer records.

3.2.4.1 Data Requirements

Each payment transaction is uniquely identified by a payment ID. Stored data includes payment ID, payment amount, deposit amount, penalty amount, payment type, payment method, receipt image and refund amount.

The payment type indicates whether the transaction is for a booking payment or a penalty charge. The payment method is QR payment. The verification status reflects whether the payment is pending, approved or rejected. Each payment record is linked to the corresponding customer and verified by an authorized admin or manager.

3.2.4.2 Transaction Requirements

Data Entry

1. Record a new payment for a booking or penalty
2. Upload payment receipt image for verification
3. Record deposit amount associated with a booking
4. Assign verifying admin/manager during payment verification

Data Update/Deletion

1. Update payment verification status whether approved or rejected
2. Update deposit balance after successful verification

Data Queries

1. Retrieve payment details for a specific booking or customer
2. Retrieve pending payments awaiting verification
3. Retrieve verified payments processed by a specific admin or manager
4. Retrieve deposit and penalty payment records
5. Check payment verification status before confirming booking
6. Generate payment summaries for reporting and auditing purposes

3.2.4.3 Local Conceptual Data Model (ERD)

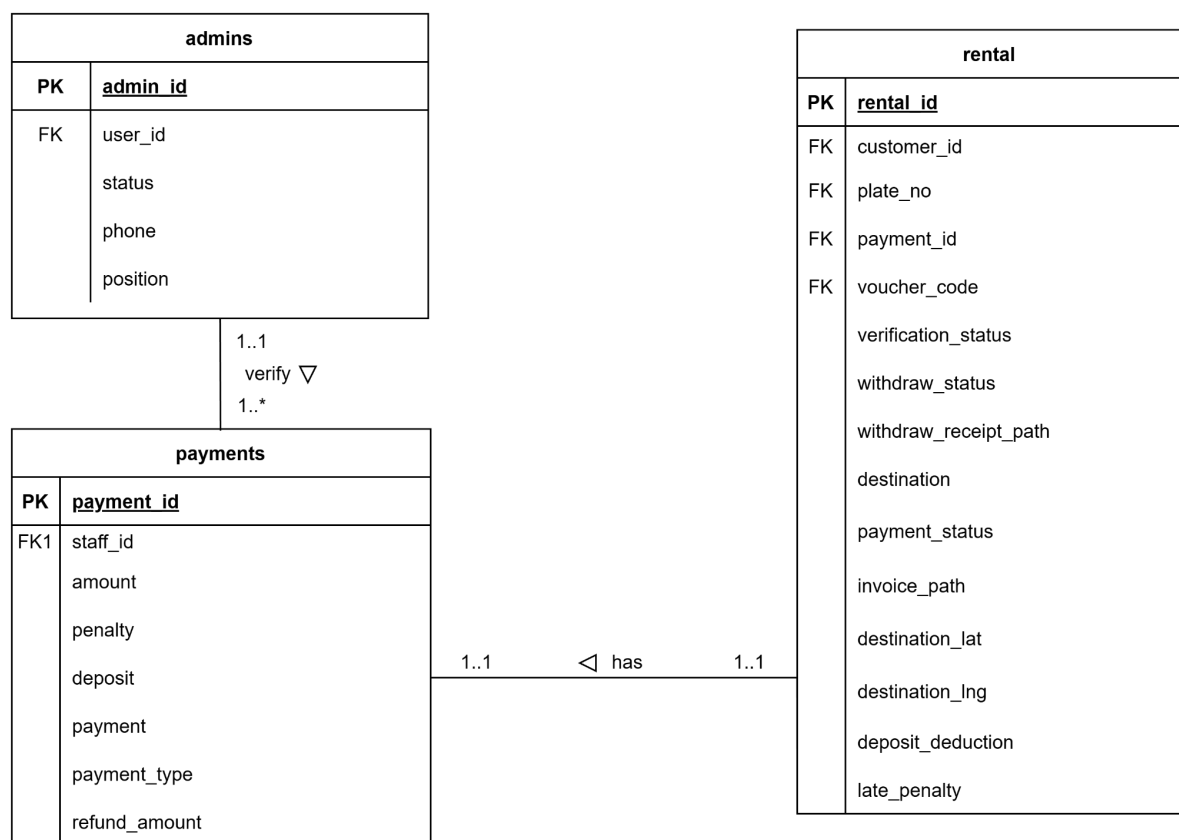


Figure 3.2.4.3.1 Local Conceptual Data Model for Payment Management Module

3.2.5 Loyalty and Rewards Module

Loyalty and Rewards Module manages customer incentive programs within the Hasta Travel & Tours Vehicle Rental Management System. It tracks customer rental activity, awards loyalty stamps and manages reward vouchers. This module aims to encourage repeat usage by rewarding loyal customers with discounts and free rental benefits.

3.2.5.1 Data Requirements

Loyalty

For the loyalty record, it is uniquely identified by a loyalty ID and is linked to a customer. Stored data includes loyalty ID, loyalty type, stamp description, rental counter, number of stamps earned.

The loyalty type indicates the nature of the reward activity such as rental-based stamps. The rental counter would track the number of qualifying rentals while the stamp count reflects accumulated loyalty progress. Loyalty records are generated automatically based on completed bookings that meet eligibility criteria or when the customers redeem the voucher.

Voucher

Each voucher is uniquely identified by a voucher ID and is linked to a customer. Stored data includes voucher ID, voucher code, voucher description, discount percentage, free rental hours, expiry date and voucher status.

Vouchers are not issued automatically. Customers may manually redeem vouchers through the system once they have sufficient loyalty stamps. Moreover, Admin and Manager can issue the vouchers as part of promotions or special cases. Voucher status indicates whether a voucher is active, used or expired.

3.2.5.2 Transaction Requirements

Data Entry

1. Generate loyalty records when a qualifying rental is completed or when a customer redeems the voucher
2. Create voucher records through customer redemption
3. Create vouchers manually by Admin or Manager to specific customers
4. Assign voucher codes and reward attributes

Data Update/Deletion

1. Update loyalty stamp count after eligible bookings
2. Increment rental counter after eligible bookings
3. Deduct rental counter upon successful redemption
4. Update voucher description, expiry date or status by Admin or Manager
5. Deactivate vouchers without deleting historical records

Data Queries

1. Retrieve loyalty stamp progress for a specific customer
2. Retrieve active, used and expired vouchers for a customer
3. Validate voucher eligibility during booking
4. Retrieve vouchers created or managed by Admin or Manager
5. Generate loyalty and voucher usage for reporting purposes

3.2.5.3 Local Conceptual Data Model (ERD)

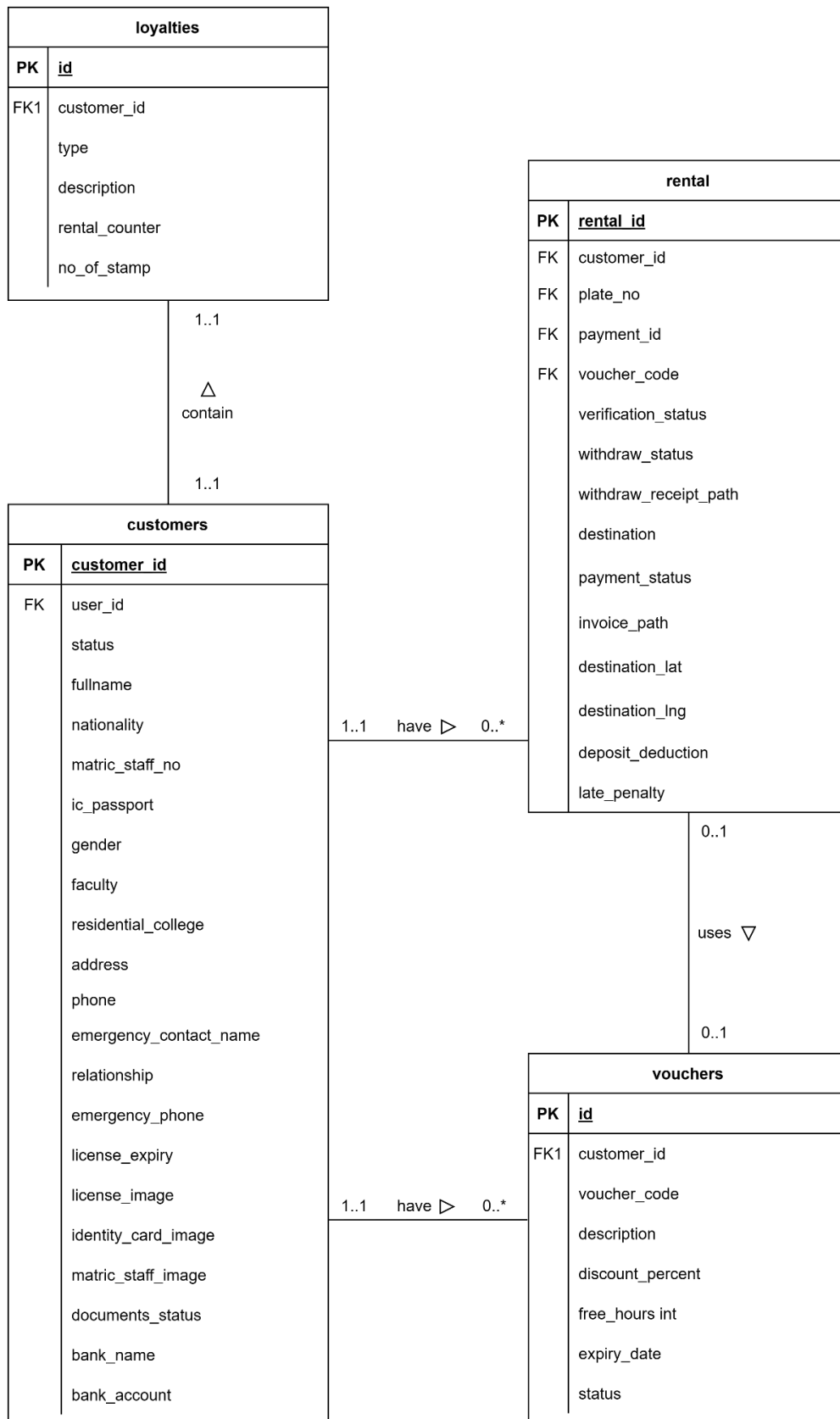


Figure 3.2.5.3.1 Local Conceptual Data Model for Loyalty and Rewards Module

3.2.6 Report and Analytics Module

The Report and Analytics Module provides administrative and managerial insights by analyzing and summarizing data collected from existing system modules. It retrieves and processes data from the User, Vehicle, Booking and Payment modules to generate meaningful reports and statistical summaries for operational monitoring and decision-making purposes.

3.2.6.1 Data Requirements

Sales and Revenue Data

Sales and revenue information is obtained from verified payment records in the Payment Module, including payment amount, deposit value, payment type and transaction timestamps. These records allow revenue to be aggregated and analyzed according to different time periods such as daily, monthly and yearly summaries. Moreover, it can be categorized based on vehicle details and customer types retrieved from the Fleet and Use Modules.

Fleet Performance

Fleet performance data is generated by analyzing vehicle availability and booking information stored in the Fleet and Booking Modules. Maintenance performance and downtime are assessed using vehicle maintenance and availability status fields. Meanwhile for the vehicle popularity metrics, they are derived from booking frequency data to identify high-demand models within the fleet.

Customer Feedback

Customer feedback data is sourced from feedback records submitted after booking completion. This includes numerical service and vehicle ratings, along with textual feedback provided by customers. Feedback data is retrieved to support trend analysis and service quality evaluation.

3.2.6.2 Transaction Requirements

Data Entry

1. Generate monthly and yearly sales reports using verified payment data
2. Generate fleet utilization analysis comparing booked hours against total available hours
3. Generate customer feedback summaries after rental completion

Data Update/Deletion

1. Regenerate reports dynamically based on updated criteria
2. Update report parameters such as data range filters to regenerate views dynamically

Data Queries

1. Retrieve total revenue for a specific date range
2. Retrieve the average vehicle rating scores
3. List all feedback entries with low ratings which is less than 3 stars for manager review
4. Export report data in PDF and Excel format for offline analysis

4.0 Database conceptual design

4.1 Conceptual ERD

Figure 4.1.1 is an example of how Hasta Travel Vehicle Management System manages information within its systems to create a data structure that is accurate and efficient by maintaining an accurate representation of the information stored on the system. Some of the basic concepts of databases have been used within this design such as entity specialization. For example, a user can act as either a customer or an admin. Defined cardinalities were created to define the relationships of connections between data elements. For instance, the customer to the rental relationship has a cardinality of 1:M. Each Rental is only associated with one vehicle, one pickup record and one return record and a payment reference to maintain transactional consistency. In addition, all of the less commonly used Supporting Entities like loyalties, payments, vehicle pictures, service records and renewal reminders have been separated into their own tables to allow for data normalization and to limit data redundancy. This ERD provides a structured, scalable model of a Vehicle Rental Company's operations in the real world and also provides for future expansion of the system.

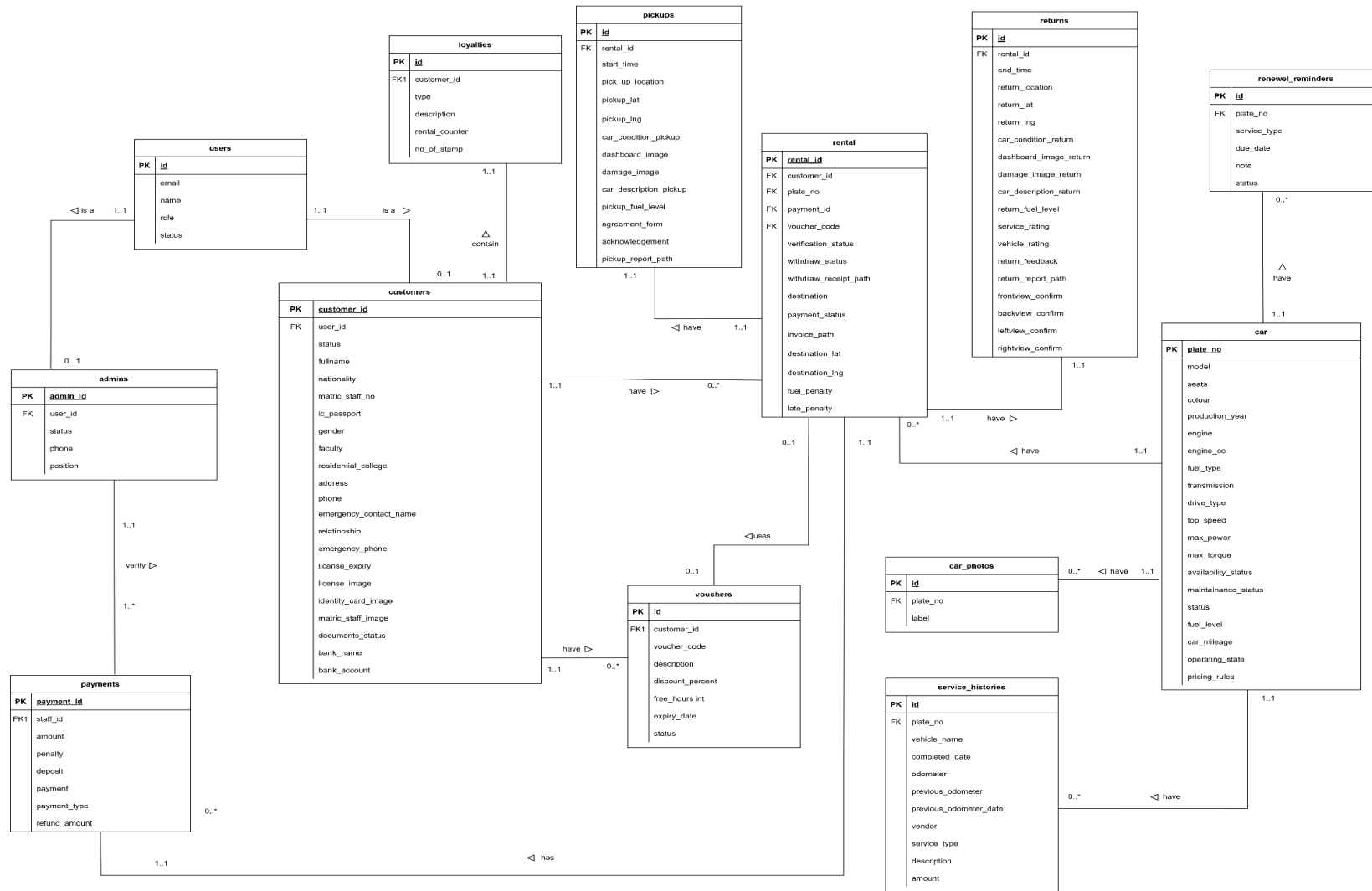


Figure 4.1.1 Entity Relationship Diagram (ERD)

4.2 Enhanced ERD (EERD)

Figure 4.2.1 builds upon the information captured in the basic ERD to provide a means of representing more advanced database concepts such as specialization, mandatory participation, and enhanced relationships. The EERD includes an example of a mandatory OR (ISA) specialization, where every user must be classified either as a customer or an admin, thus enforcing role integrity at the database level. The rental entity is identified as the primary transaction and is mandatorily related to both pickup and return entities to provide for a complete rental lifecycle. The additional entities such as vouchers that have a voucher redemption relationship, loyalties, payments, car photos, service histories, and renewal reminders, have been separated to allow for normalization of data within the database and to avoid redundant data. The overall result of the EERD is a more expressive and accurate database design that captures business rules, business constraints, and business logic more accurately than a basic ERD

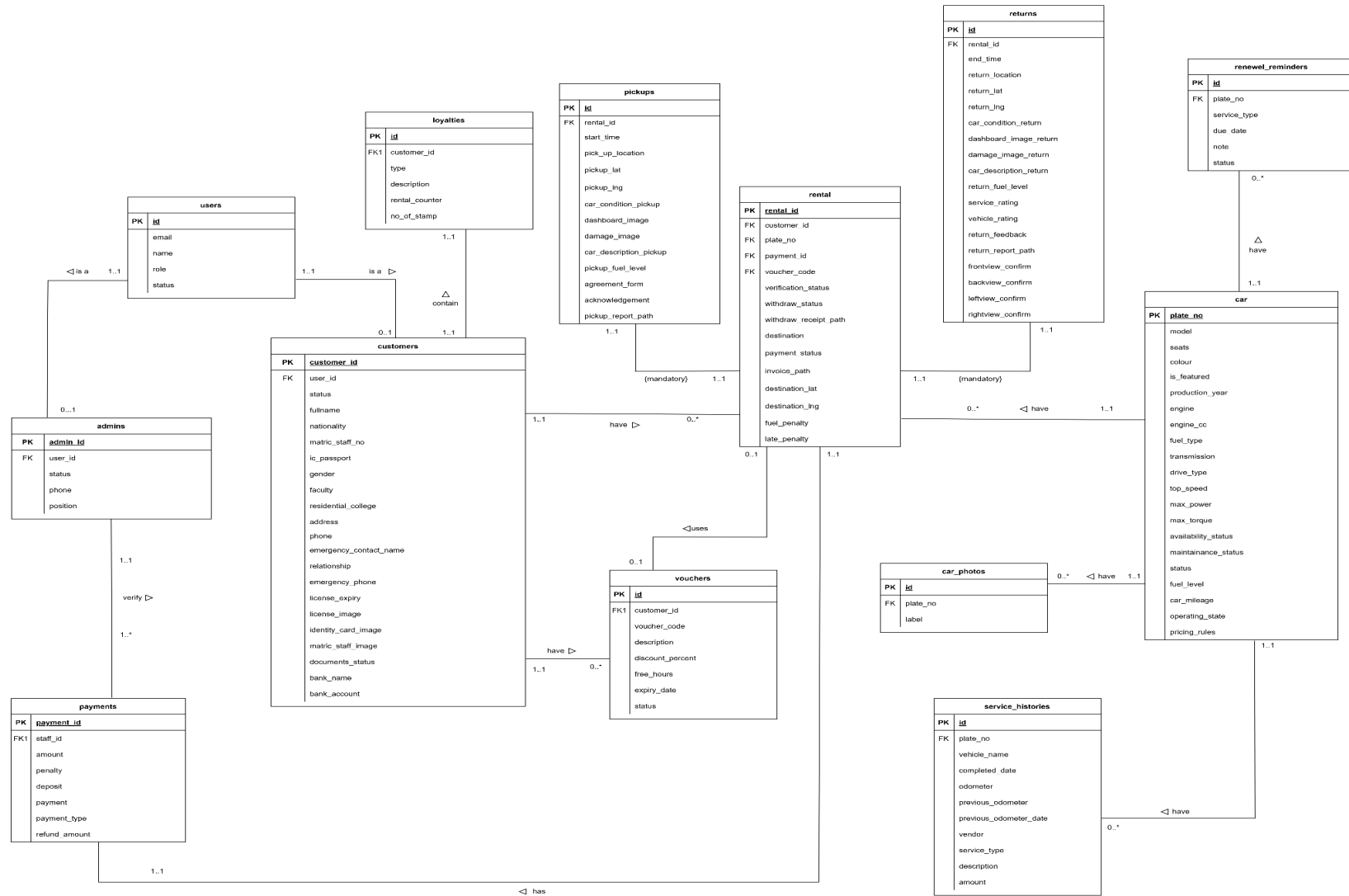


Figure 4.2.1 Enhanced Entity Relationship Diagram (EERD)

5.0 Data Dictionary

1. admins

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
admin_id	varchar	50	Unique admin identifier	-	Primary Key	Yes
user_id	bigint	-	Unique user identifier	-	Foreign Key => user.id	Yes
status	varchar	20	Current status of the admin account	active/deactivated	-	Yes
phone	varchar	20	Admin's contact phone number	-	-	No
position	varchar	20	Admin's job position	-	-	No

2. car

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
plate_no	varchar	20	Vehicle's plate number	Alphanumeric	Primary Key	Yes
model	varchar	150	Vehicle's model	-	-	Yes
seats	tinyint	-	Number of seats	1 – 9	-	No

colour	varchar	30	Vehicle colour	-	-	No
is_featured	tinyint	1	Featured vehicle flag	0 = No, 1 = Yes	-	Yes
availability_status	tinyint	1	Availability of vehicle	available / unavailable	-	Yes
status	varchar	20	Operational status of car	active / retired	-	Yes
fuel_level	integer	-	The fuel level for the vehicle	-	-	Yes
car_mileage	decimal	10, 2	Vehicle's mileage	≥ 0	-	Yes
production_year	year	-	Year of manufacture	YYYY	-	No
engine	varchar	100	Engine description	-	-	No
engine_cc	integer	-	Engine capacity(cc)	-	-	No
fuel_type	varchar	20	Type of fuel	Petrol/Diesel/EV	-	No
transmission	varchar	20	Transmission type	Automatic / Manual	-	No
drive_type	varchar	20	Drive type	FWD/RWD/AWD	-	No
top_speed	integer	-	Maximum speed (km/h)	-	-	No

max_power	decimal	8, 2	Maximum power (HP)	-	-	No
max_torque	decimal	8, 2	Maximum torque (Nm)	-	-	No
maintenance_status	tinyint	1	Maintenance state	0 = OK, 1 = Under Maintenance	-	Yes
operating_state	varchar	50	State where vehicle operates	-	-	No
pricing_rules	json	-	Hourly pricing rates of the vehicle	-	-	Yes

3. car_photos

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint (UNSIGNED)	-	Unique identifier for each vehicle photo record	Auto-increment integer	Primary Key	Yes
plate_no	varchar	20	Vehicle's plate number	Alphanumeric	Foreign key => car.plate_no	Yes
label	varchar	50	Label of the image	Front, Back, Left, Right, Interior 1,	-	No

				Interior 2		
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4. customers

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
customer_id	varchar	50	Unique customer identifier	-	Primary key	Yes
user_id	bigint	-	Unique user identifier	-	Foreign Key => user.id	Yes
status	varchar	20	Current status of the customer account	active/ blacklisted	-	Yes
fullname	varchar	100	Customer's full name as per IC/passport	-	-	No
nationality	varchar	100	Customer's nationality	-	-	No
matric_staff_no	varchar	20	Student or staff matric number	-	Unique key	No
ic_passport	varchar	15	IC number (local) or passport number (foreign)	-	-	No
gender	varchar	50	Customer's gender	"Male", "Female", "Prefer not to	-	No

				say”		
faculty	varchar	100	Customer’s faculty	-	-	No
residential_coll ege	varchar	100	Customer’s residential college	-	-	No
address	text	-	Customer’s home address	-	-	No
phone	varchar	20	Customer’s contact phone number	Digits, ‘+’ allowed	Unique key	No
emergency_con tact_name	varchar	100	Name of emergency contact person	-	-	No
relationship	varchar	50	Relationship between customer and emergency contact person	-	-	No
emergency_pho ne	varchar	20	Emergency contact phone number	Digits ‘+’ allowed	-	No
license_expiry	date	-	Driving license expiry date	-	-	No
license_image	varchar	255	File path or URL to driving license image	-	-	No
identity_card_i	varchar	255	File path or URL to identity	-	-	No

mage			card image			
matric_staff_image	varchar	255	File path or URL to matric card image	-	-	No
documents_status	enum	-	Verification status of submitted documents	'pending', 'approved', 'rejected'		No
bank_name	varchar	100	Customer's bank name	-	-	No
bank_account	varchar	30	Customer's bank account number	-	-	No

5. loyalties

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint	-	Unique loyalty identifier	-	Primary Key	Yes
customer_id	varchar	50	ID of the customer who owns this loyalty record	-	Foreign Key -> customer.customer_id	Yes
type	varchar	50	Indicates if stamp was earned or redeemed	'earned', 'redeemed'	-	Yes

description	varchar	255	Description of the loyalty action	-	-	Yes
rental_counter	integer	-	Current Redeemable Stamps	0 or above	-	Yes
no_of_stamp	integer	-	Lifetime Total Stamps	0 or above	-	Yes

6. payments

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
payment_id	varchar	20	Unique payment identifier	-	Primary Key	Yes
staff_id	bigint	50	Unique staff identifier	-	Foreign Key -> admins.admin_id	No
amount	decimal	10, 2	Amount need to pay	-	-	Yes
penalty	decimal	10, 2	Penalty amount that needs to pay	-	-	Yes
deposit	decimal	10, 2	Deposit amount that needs to pay	-	-	Yes
payment	varchar	20	Payment method to choose	Valid bank QR	-	Yes
payment_type	varchar	20	Reason of payment whether	booking /	-	Yes

			for rental or penalty	penalty		
refund_amount	decimal	10, 2	The amount that needed to be refunded to the customer	-	-	Yes

7. renewal_reminders

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint (UNSIGNED)	-	Unique identifier for each renewal reminder record	Auto-increment integer	Primary Key	Yes
plate_no	varchar	20	Vehicle plate number associated with the reminder	Alphanumeric	Foreign key => car.plate_no	Yes
service_type	enum	-	Type of renewal or service reminder	Insurance, Investor Commission, Monthly Installment, Roadtax, Others	-	Yes
due_date	date	-	Due date for the renewal or service	Valid date	-	Yes
note	text	-	Additional remarks or notes for the	Free text	-	

			reminder			
status	enum	-	Current reminder status	upcoming, notified, completed	-	Yes

8. rental

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
rental_id	bigint	-	Unique identifier for each rental transaction	Auto-increment positive integer	Primary Key	Yes
customer_id	varchar	50	Identifier of the customer who made the rental	Existing customer IDs	Foreign Key -> customer.customer_id	Yes
plate_no	varchar	10	Vehicle plate number rented	Registered vehicle plate numbers	Foreign Key -> car.plate_no	Yes
payment_id	varchar	50	Unique payment id	System-generated payment IDs	Foreign Key -> payment.payment_id	Yes
voucher_code	varchar	50	Voucher code applied to rental	Existing voucher	Foreign Key -> vouchers.voucher_code	No

				codes	de	
verification_status	varchar	50	Status of verification	pending/ verified/ done/ renting/ Cancelled/ failed	-	Yes
withdraw_status	varchar	20	Status of deposit withdrawal/refund	pending/ done	-	Yes
withdraw_receipt_path	varchar	255	The receipt for withdraw reposit that upload by admin	Valid file path	-	No
destination	varchar	255	Destination entered by customer	Free text location	-	Yes
payment_status	varchar	50	The status to record the payment of user	pending/ paid/ failed/ cancelled	-	No
invoice_path	varchar	255	The autogenerated invoice of this rental	Valid file path	-	No
destination_lat	decimal	10,7	Latitude of destination location	Valid latitude coordinate	-	Yes

destination_lng	decimal	10,7	Longitude of destination location	Valid longitude coordinate	-	Yes
fuel_penalty	decimal	8, 2	Charges incurred when fuel level during return is less than pickup	-	-	
late_penalty	decimal	10,2	Charges incurred due to late return	-	-	Yes

9. pickups

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint	-	Unique booking identifier	Auto-increment	Primary Key	Yes
rental_id	bigint	-	References the rental associated with this pickup	Existing rental.rental_id	Foreign Key-> rental.rental_id	Yes
start_time	datetime	-	Scheduled pickup start time and date	Valid datetime	-	Yes
pick_up_location	varchar	255	The pickup location	Text	-	Yes
pickup_lat	decimal	10,7	Latitude coordinate of pickup location	Any valid latitude	-	No

pickup_lng	decimal	10,7	Longitude coordinate of pickup location	Any valid longitude	-	No
car_condition_pickup	text	-	Text describing vehicle condition at pickup	Text	-	No
dashboard_image	varchar	255	Image path of vehicle dashboard at pickup	File path	-	No
damage_image	varchar	255	Image path showing any vehicle damage at pickup	File path	-	No
car_description_pickup	text	-	Additional notes describing vehicle condition at pickup	Text	-	No
pickup_fuel_level	integer	1	Fuel level at pickup	1 to 8	-	No
agreement_form	varchar	255	Signed rental agreement file path	File path	-	No
acknowledgement	tinyint	1	Customer acknowledgement of pickup	0,1	-	Yes
pickup_report_path	varchar	255	Generated pickup inspection report file path	File path	-	No

10. returns

Column	Data Type	Length	Description	Allowed	Relationship	Required
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				Values		
id	bigint	-	Unique identifier for return record	Auto-increment	Primary Key	Yes
rental_id	bigint	-	References the rental associated with this return	Existing rental.rental_id	Foreign Key -> customer.customer_id	Yes
end_time	datetime	-	Actual return date and time of the vehicle	Valid datetime	-	Yes
return_location	varchar	255	Vehicle return location name or address	Text	-	Yes
return_lat	decimal	10,7	Latitude of return location	Any valid latitude	-	No
return_lng	decimal	10,7	Longitude of return location	Any valid longitude	-	No
car_condition_return	text	-	Text describing vehicle condition at return	Text	-	No
dashboard_image_return	varchar	255	Image path of dashboard at return	File path	-	No
dashboard_image_return	varchar	255	Image path showing vehicle damage at return	File path	-	No

car_description_return	text	-	Additional notes on vehicle condition during return	text	-	No
return_fuel_level	int	-	Fuel level at time of return	0-8	-	No
service_rating	int	-	Customer rating for rental service	1-5	-	No
vehicle_rating	int	-	Customer rating for vehicle condition	1-5	-	No
return_feedback	text	-	Customer feedback or comments after return	text	-	No
return_report_path	varchar	255	Generated return inspection report file path	File path	-	No
frontview_confirmation	tinyint	1	Confirmation that front view inspection is completed	0/1	-	Yes
backview_confirmation	tinyint	1	Confirmation that back view inspection is completed	0/1	-	Yes
leftview_confirmation	tinyint	1	Confirmation that left view inspection is completed	0/1	-	Yes
rightview_confirmation	tinyint	1	Confirmation that right view inspection is completed	0/1	-	Yes

11. service_histories

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint (UNSIGNED)	-	Unique identifier for service record	Auto-increment integer	Primary Key	Yes
plate_no	varchar	20	Vehicle plate number associated with the service	Alphanumeric	Foreign key => car.plate_no	Yes
vehicle_name	varchar	100	Vehicle model/name	Text	-	Yes
completed_date	date	-	Date service was completed	Valid date	-	Yes
odometer	integer	-	Vehicle mileage at service time (km)	≥ 0	-	No
previous_odometer	integer	-	Previous recorded mileage	≥ 0	-	No
previous_odometer_date	date	-	Date of previous mileage record	Valid date	-	No

vendor	varchar	100	Service provider / workshop	Text	-	No
service_type	varchar	50	Type of service	Oil Change, Meter Update, Repair, Inspection	-	No
description	text	-	Additional service details	Free text	-	No
amount	decimal	10,2	Service cost (MYR)	≥ 0.00	-	Yes

12. users

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	varchar	-	Unique user identifier used for authentication and internal references	-	Primary key	yes
email	varchar	100	User's login email address	Valid email format	Unique key	yes
role	varchar	50	Defines the role of the user in the system	customer/ admin/ manager	-	yes
name	varchar	50	Username use in system	-	-	Yes

status	varchar	20	account status	active / deactivate	-	Yes
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13. vouchers

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint	-	Unique identifier for the voucher	-	Primary key	Yes
customer_id	varchar	50	ID of the customer who owns the voucher	-	Foreign Key -> customer.customer_id	Yes
voucher_code	varchar	50	Unique code for each voucher	-	Unique Key	Yes
description	string	255	Detail about the voucher	-	-	No
discount_percent	integer	-	Percentage discount amount	-	-	No
free_hours	integer	-	Free rental hours granted	-	-	No
expiry_date	date	-	The date when the voucher expires	-	-	Yes

status	varchar	50	Current state of the voucher	'active', 'used', 'expired'	-	Yes
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6.0 Summary

The proposed system for Hasta Travel and Tours represents a strategic shift from traditional manual operations to a comprehensive digital platform designed to optimize the vehicle rental booking process. This transformation is essential for unlocking the company's full business potential while eliminating the inefficiencies that currently compromise productivity and service quality. By incorporating advanced automation, intuitive design, and streamlined workflows, the system is positioned to deliver a superior experience for both customers and internal stakeholders.

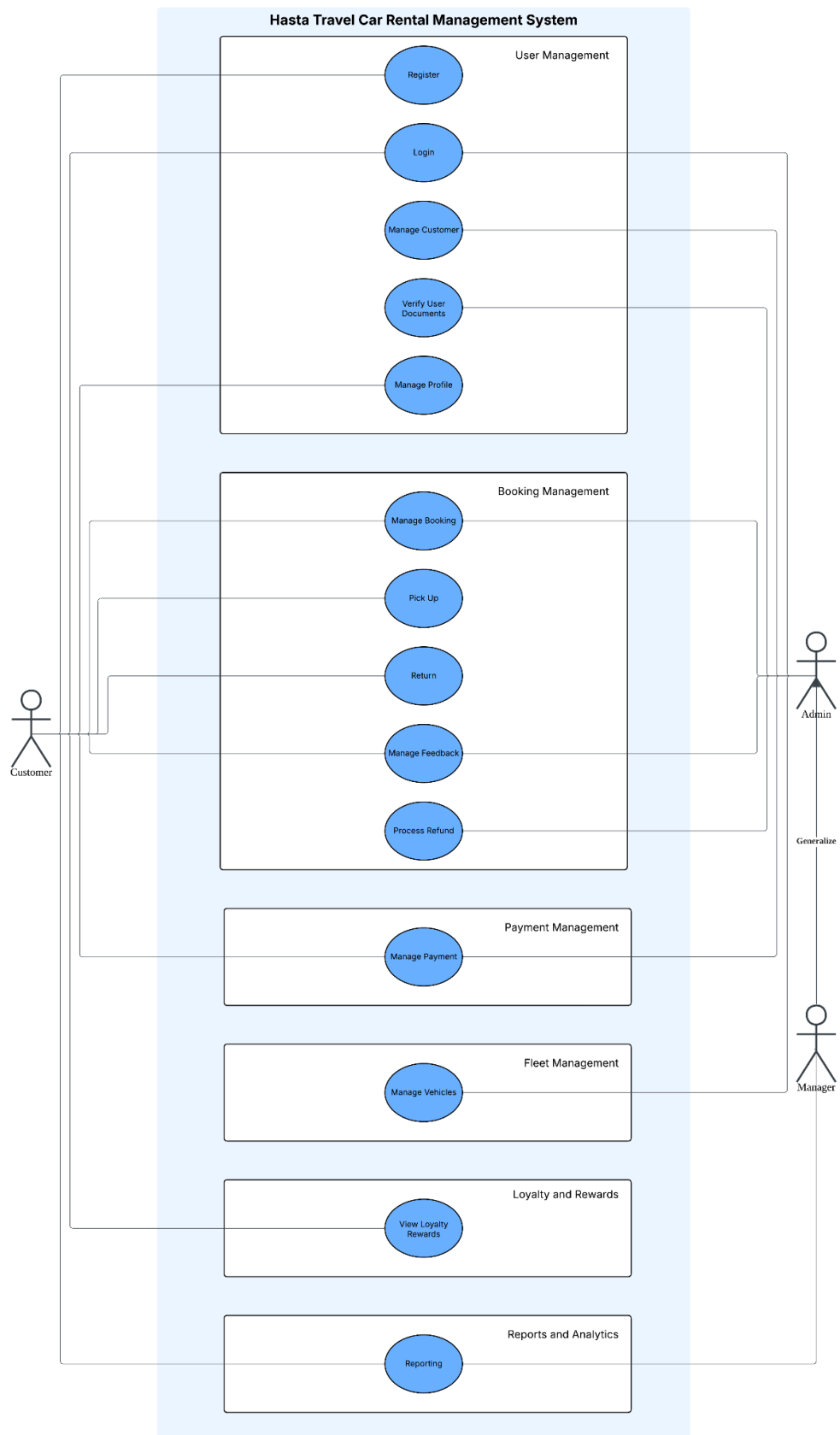
Central to the system are features that prioritize convenience and accessibility. Customers will benefit from a dynamic online booking form that displays real-time vehicle availability and pricing, complemented by automated notifications and instant e-receipt generation. The mobile-responsive interface ensures that users can complete transactions quickly and effortlessly from any device, fostering greater confidence and satisfaction throughout the booking journey.

From an administrative perspective, the system strengthens data accuracy and accountability by automating essential functions such as payment verification, vehicle condition documentation, and revenue tracking. Real-time data synchronization minimizes the likelihood of human error while providing transparent access to financial and rental records. Staff and management gain enhanced operational oversight through organized digital reports and centralized documentation, enabling more informed decision-making.

Ultimately, this system establishes a robust foundation for operational excellence by integrating efficiency, reliability, and transparency into every aspect of the business. The accompanying diagrams and documentation within this proposal provide a detailed blueprint of the database architecture, offering clear insight into the processes, entities, and attributes that underpin the Hasta Travel System.

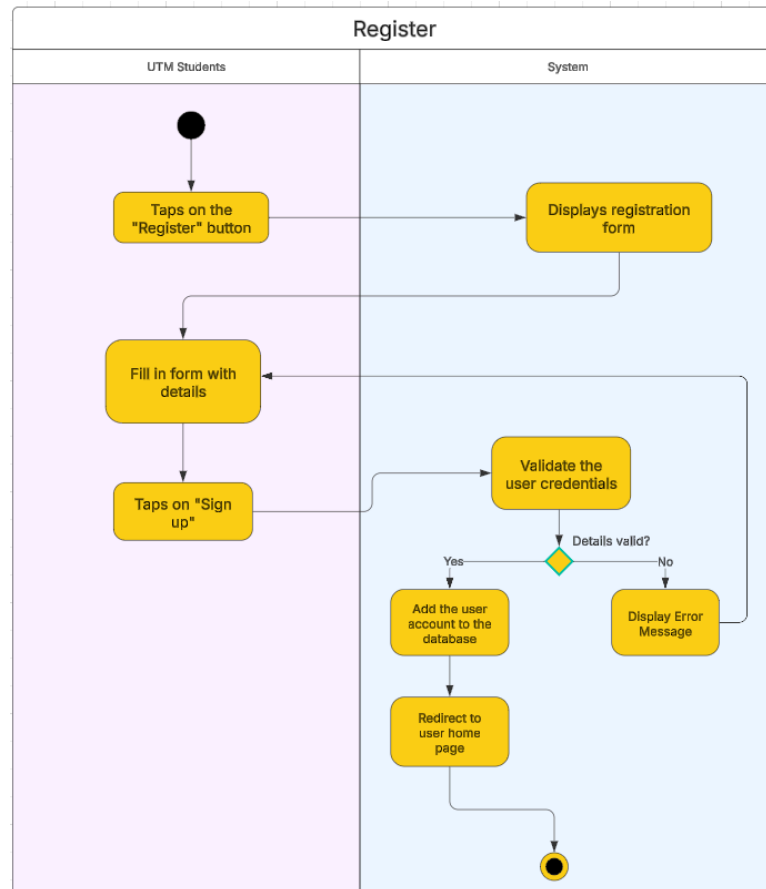
7.0 Appendices

7.1 Use Case Diagram

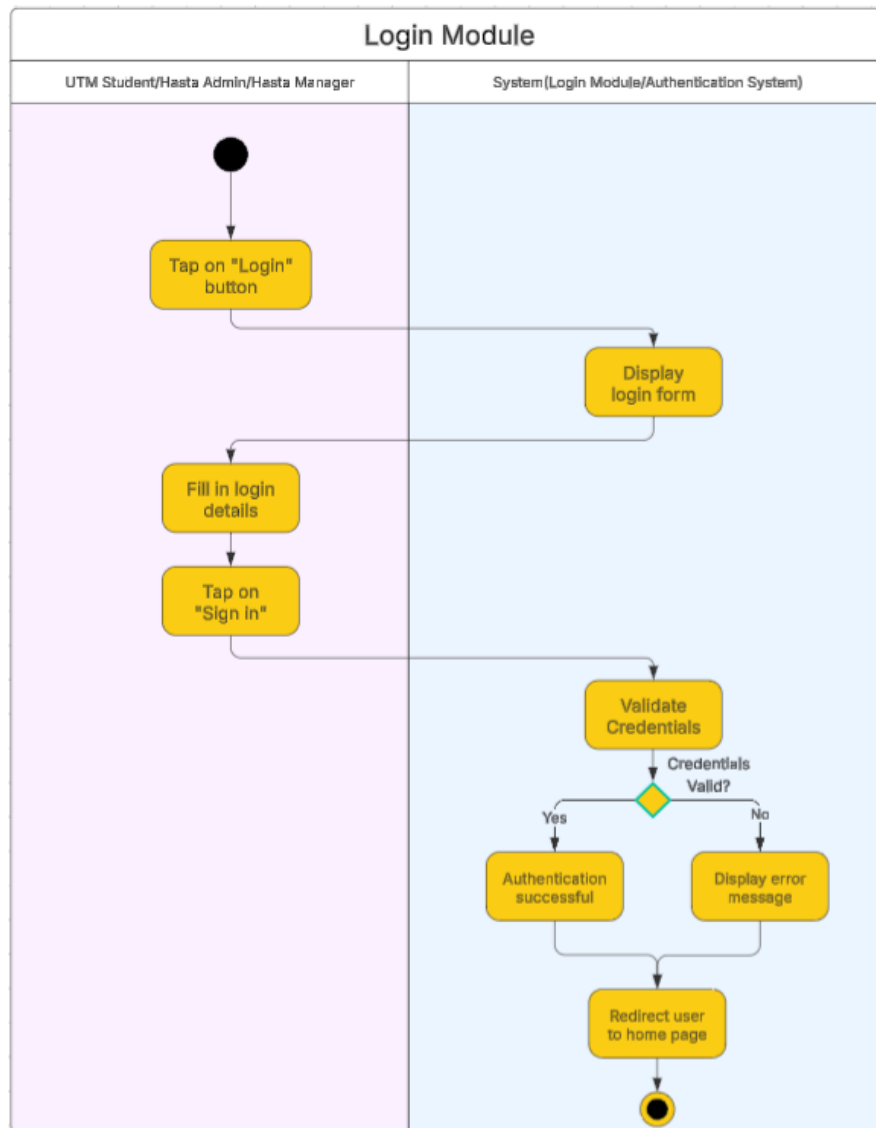


7.2 Activity Diagram

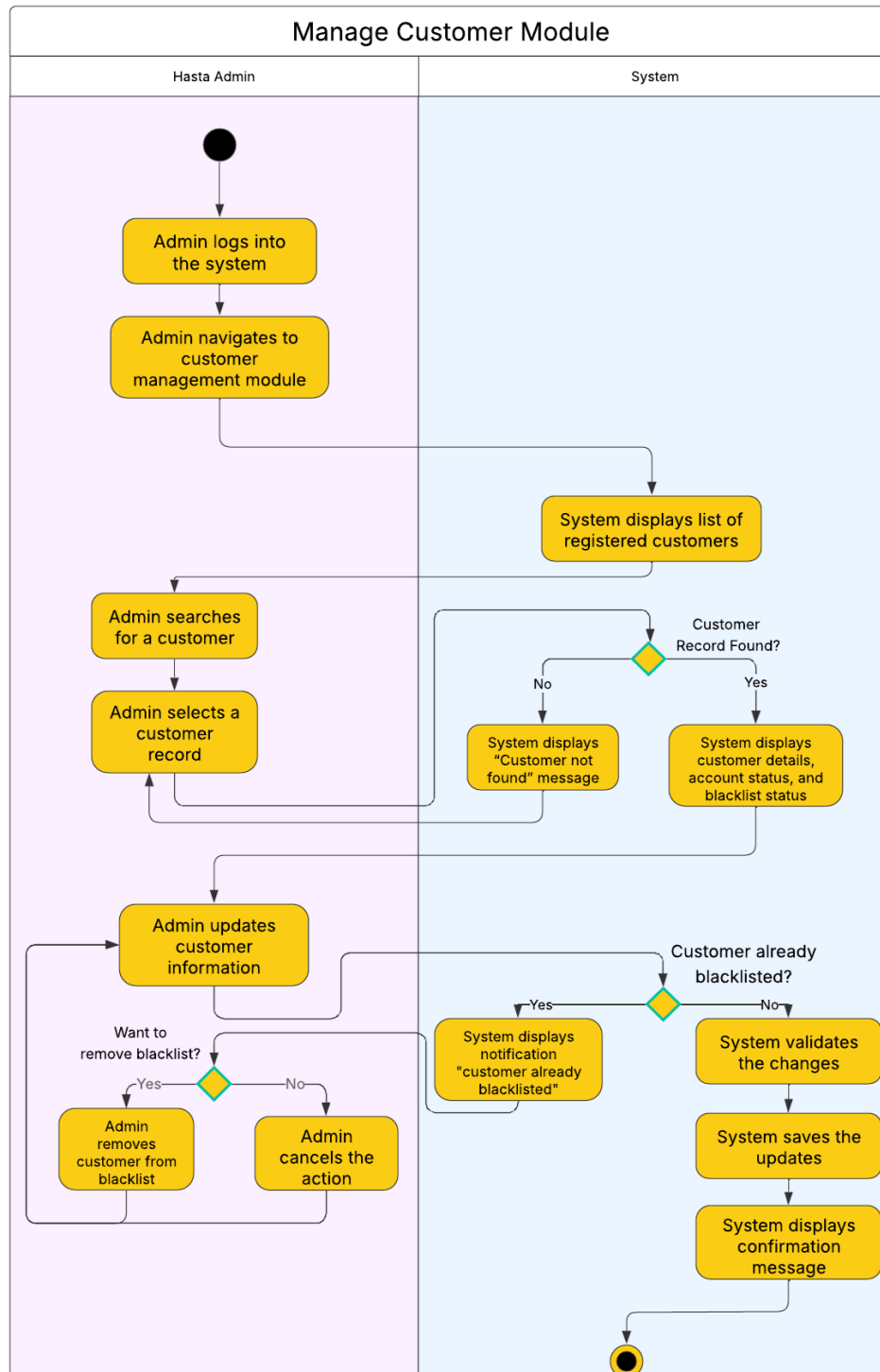
1. Register



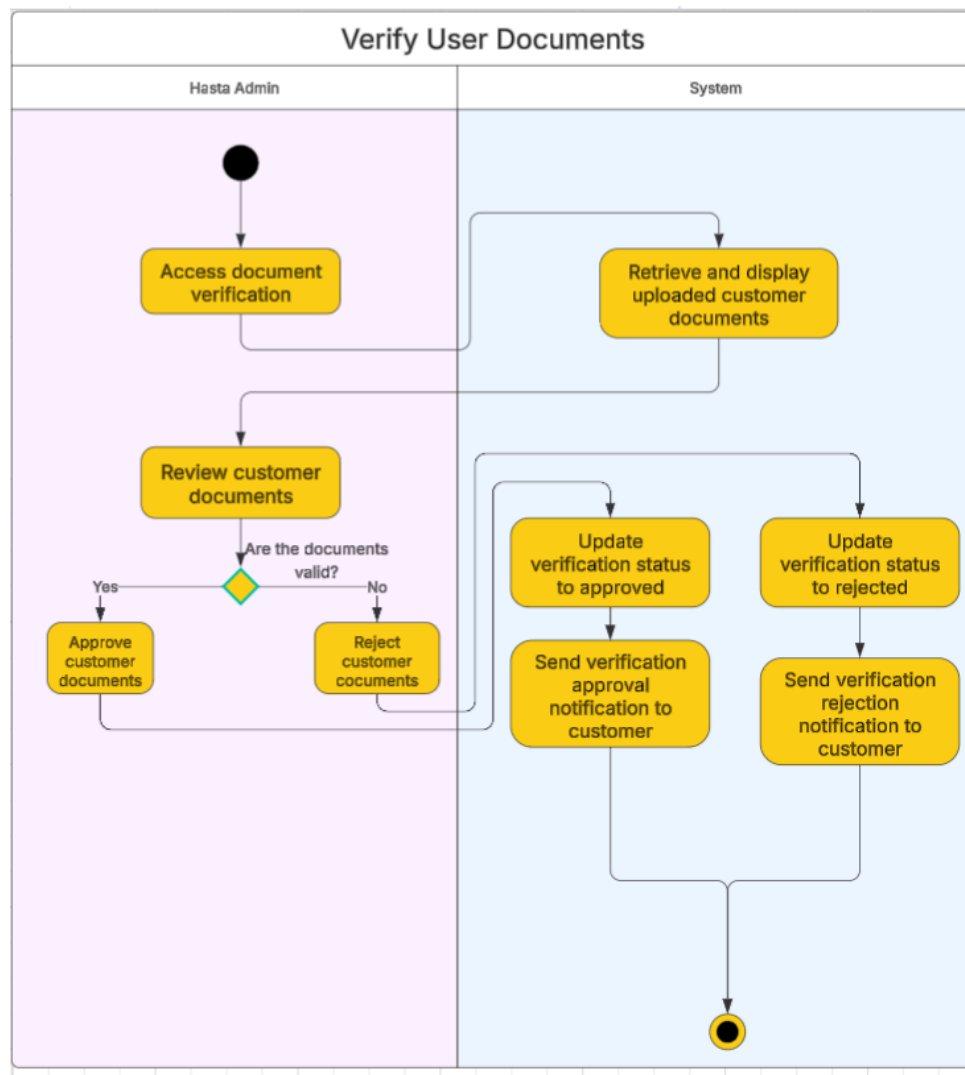
2. Login



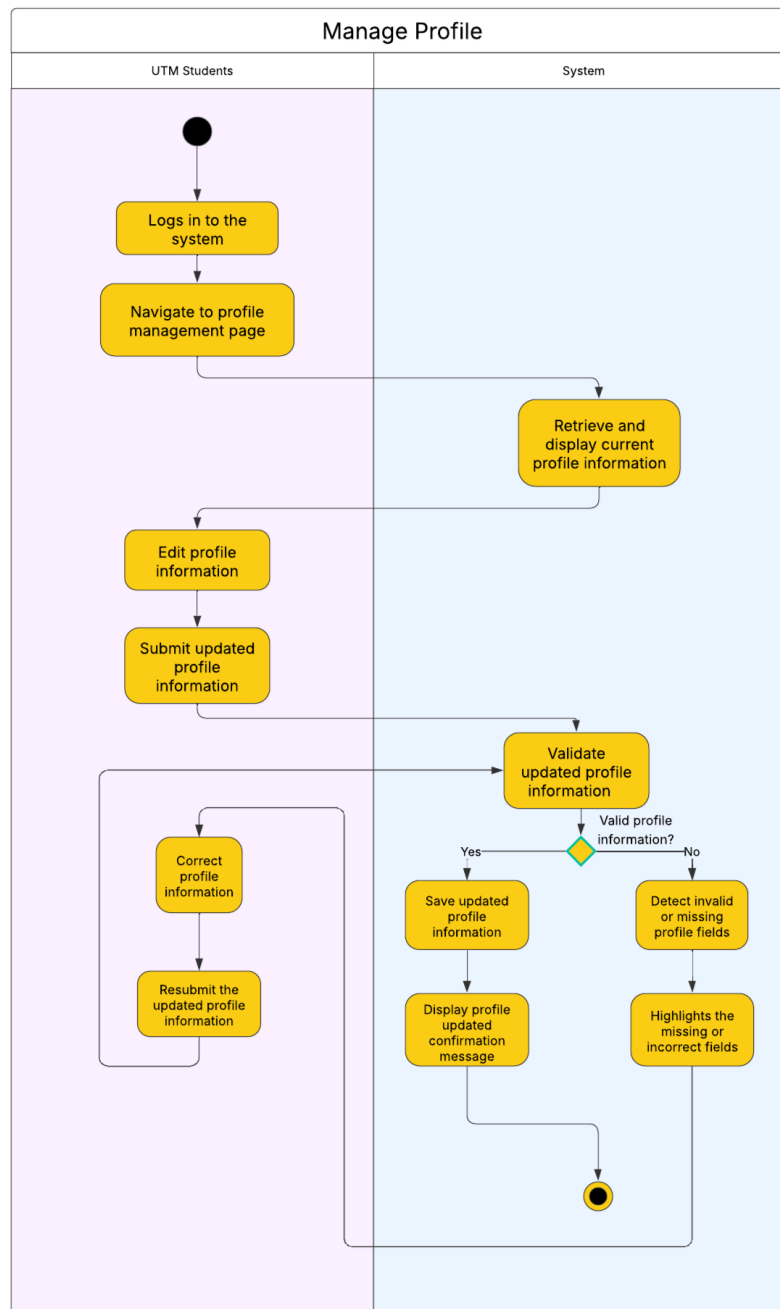
3. Manage Customer



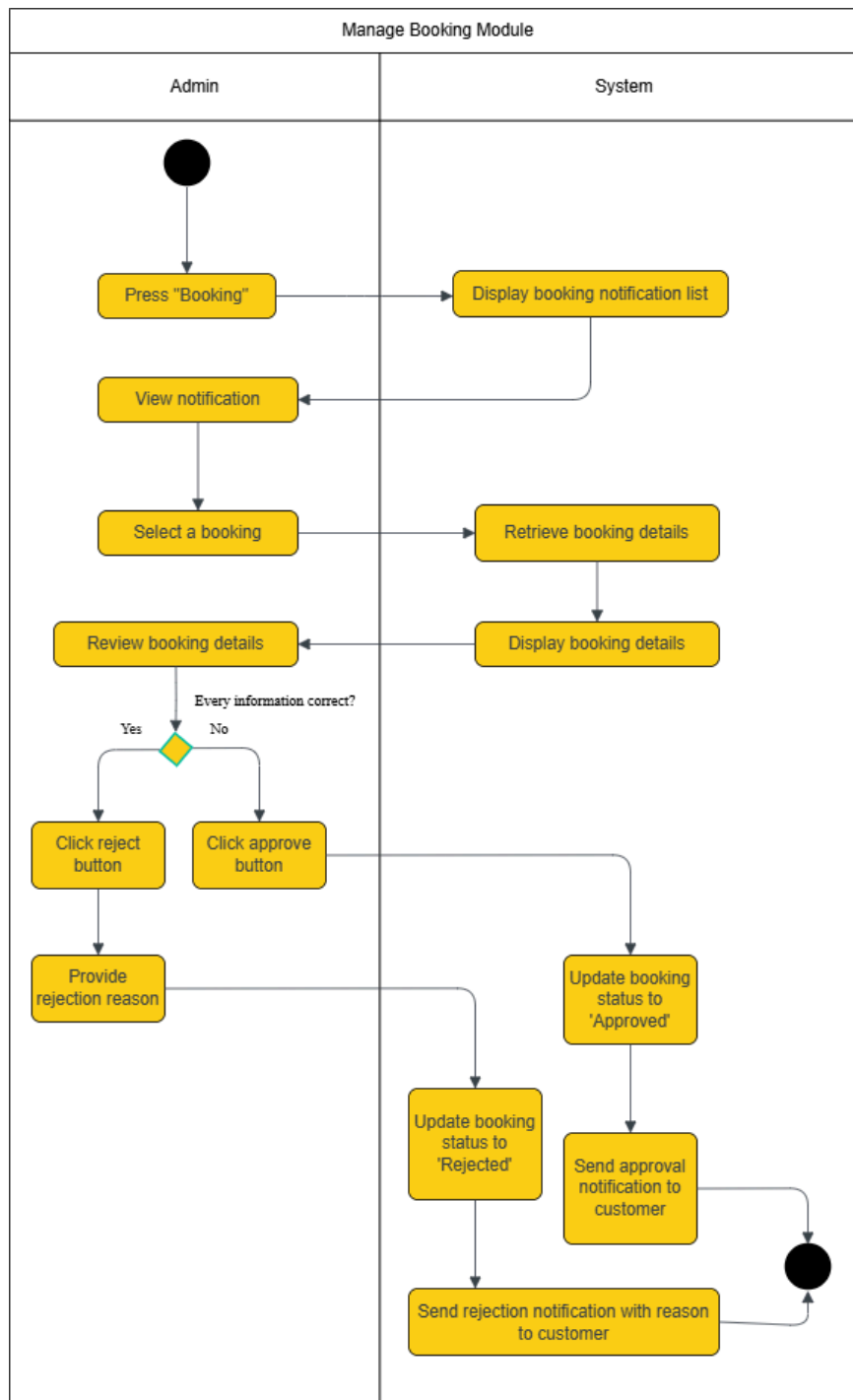
4. Verify User Documents



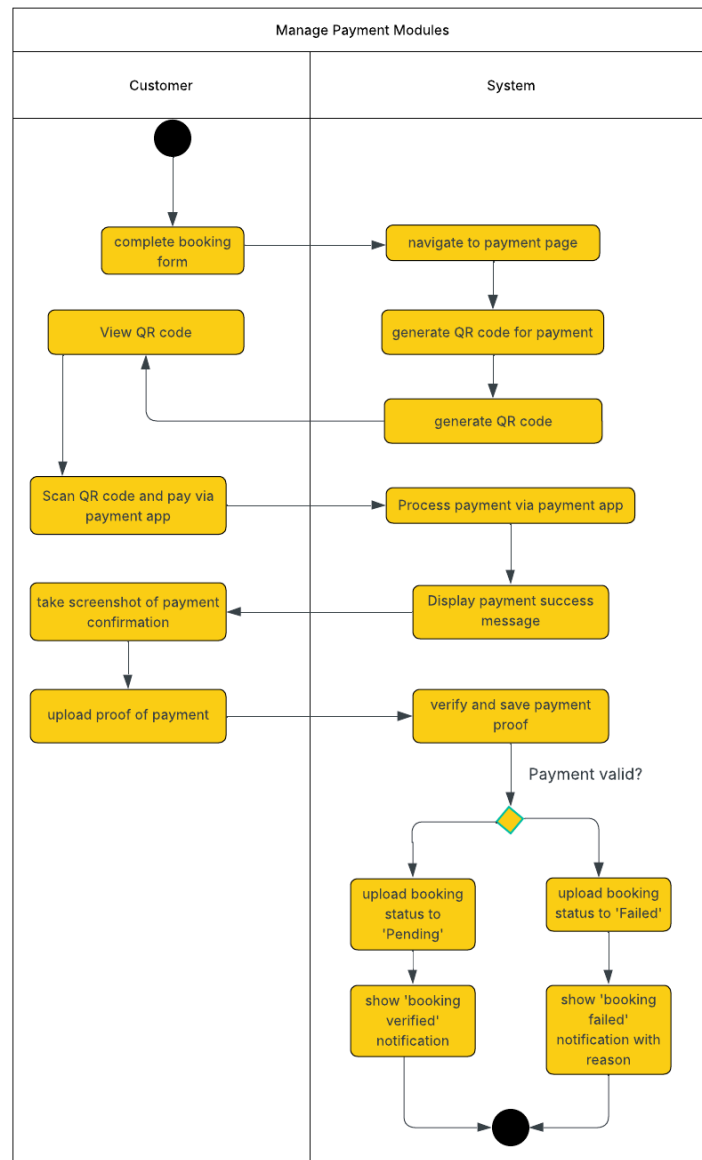
5. Manage Profile

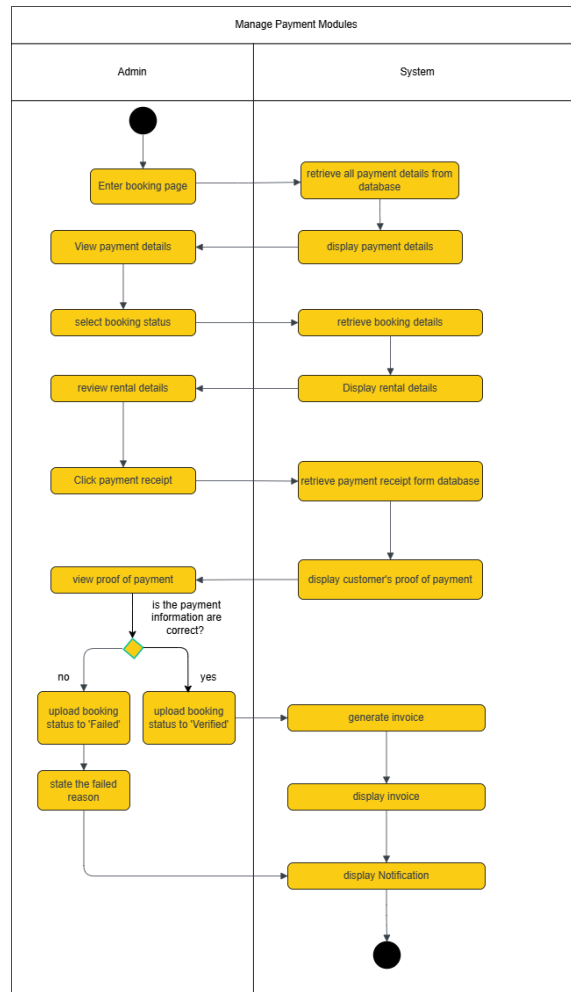


6. Manage Booking

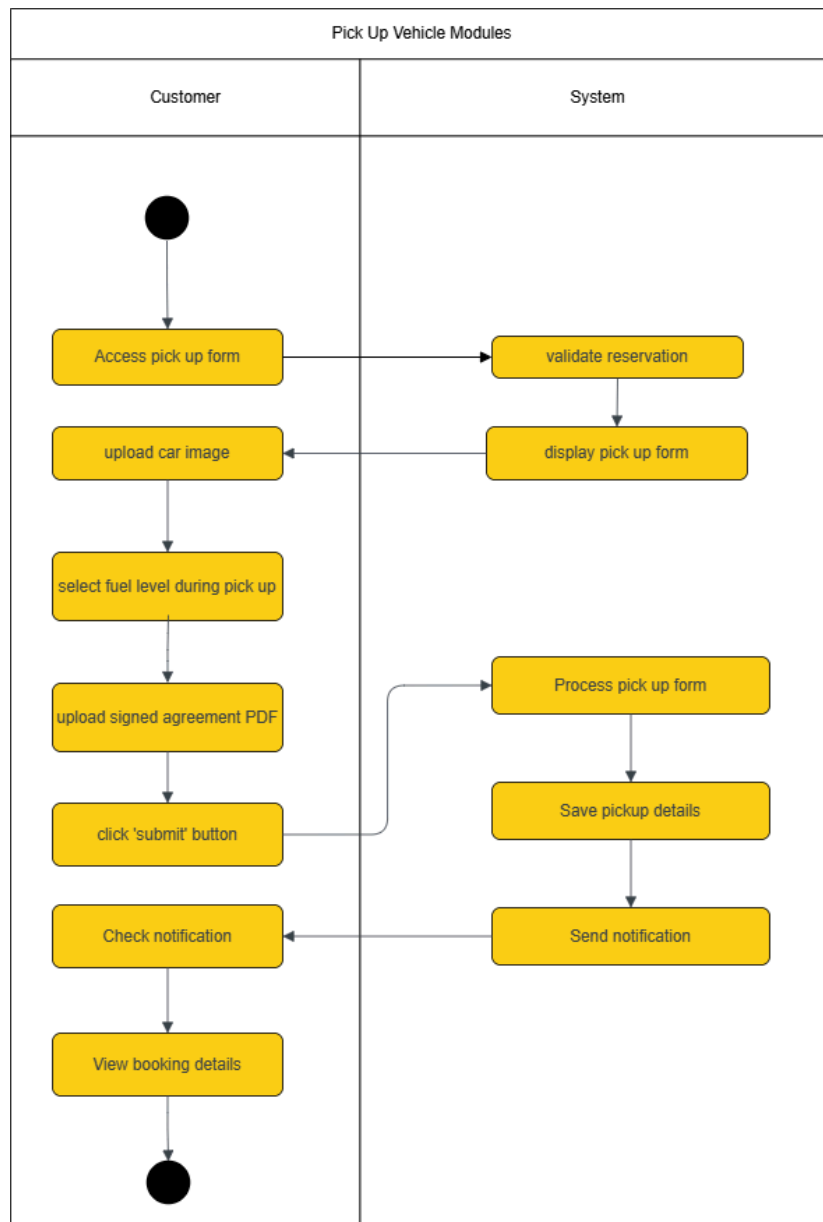


7. Manage Payment

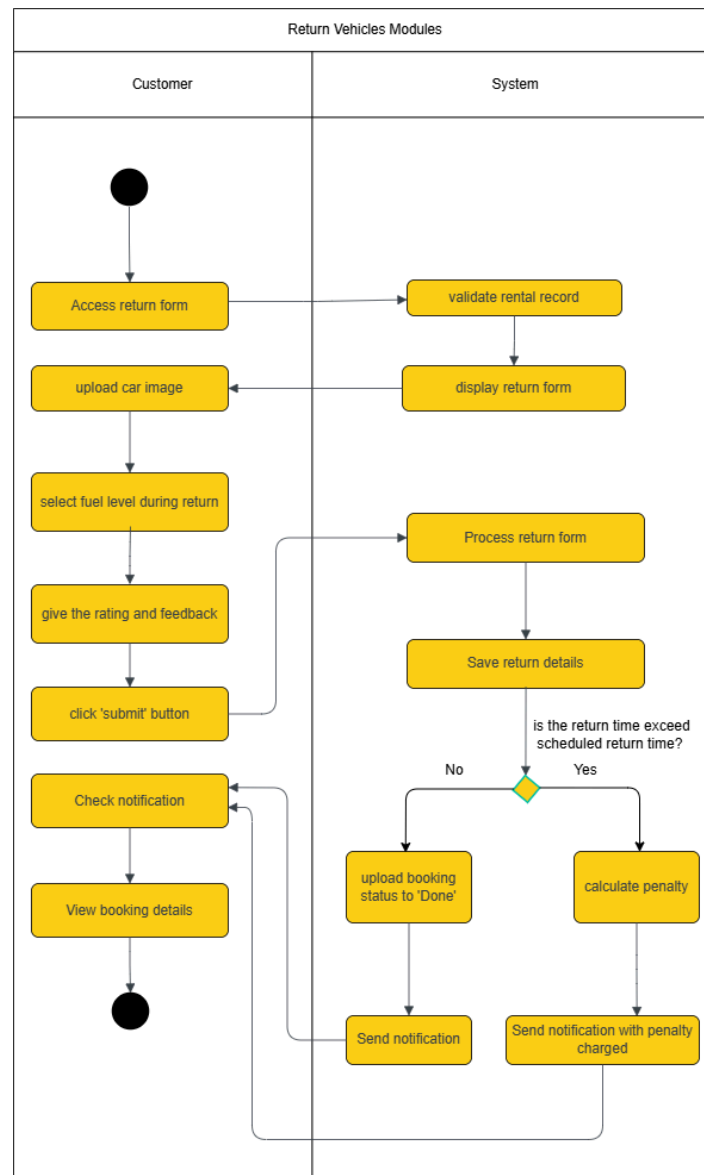




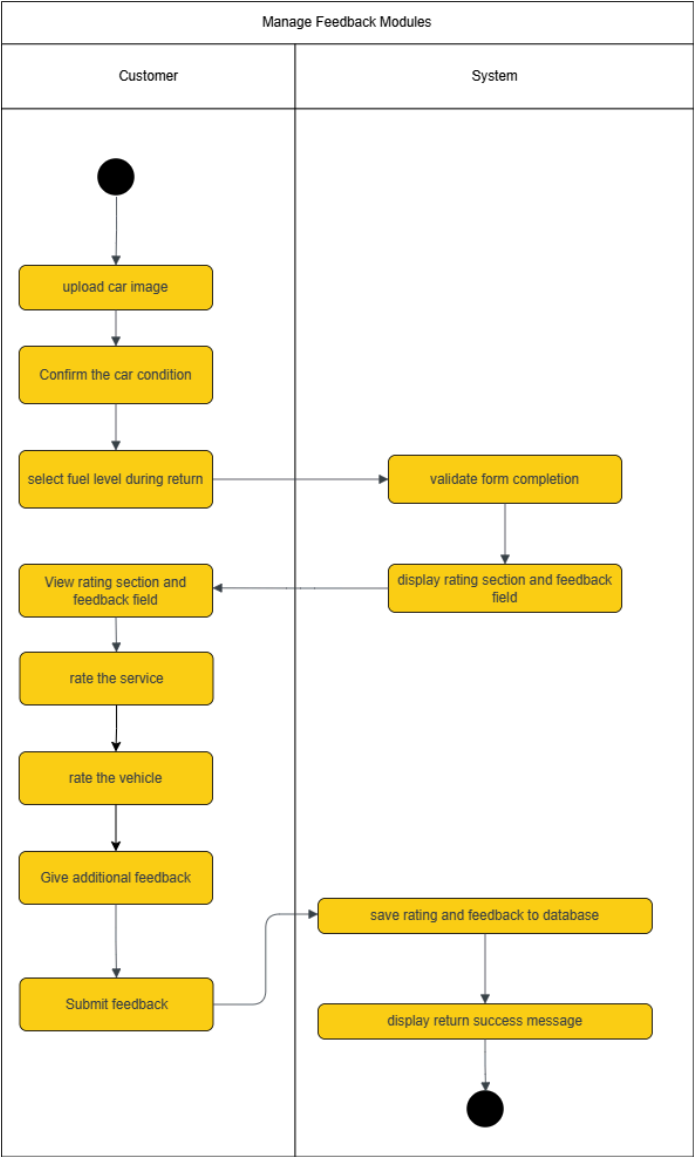
8. Pick Up Vehicle

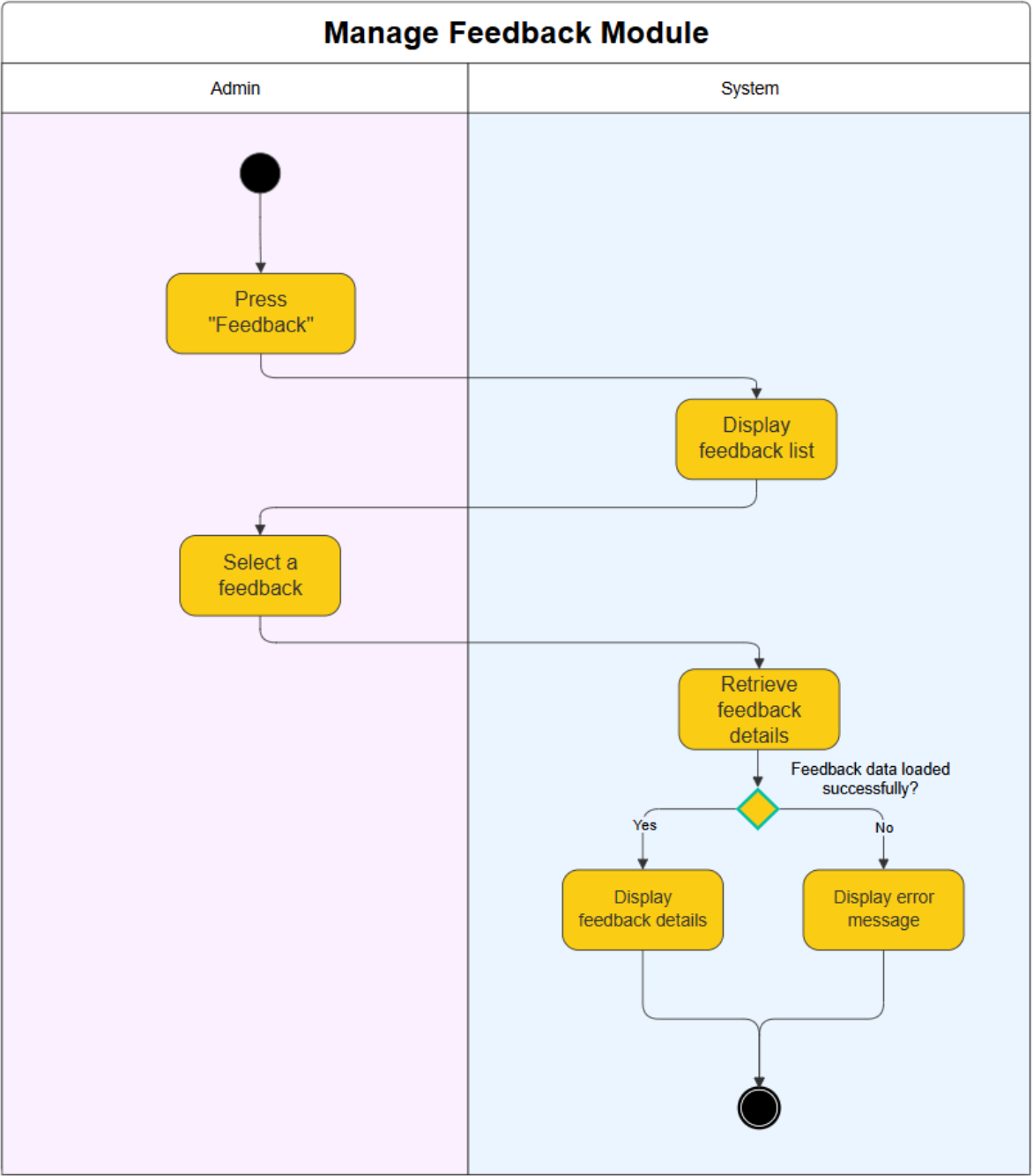


9. Return Vehicle

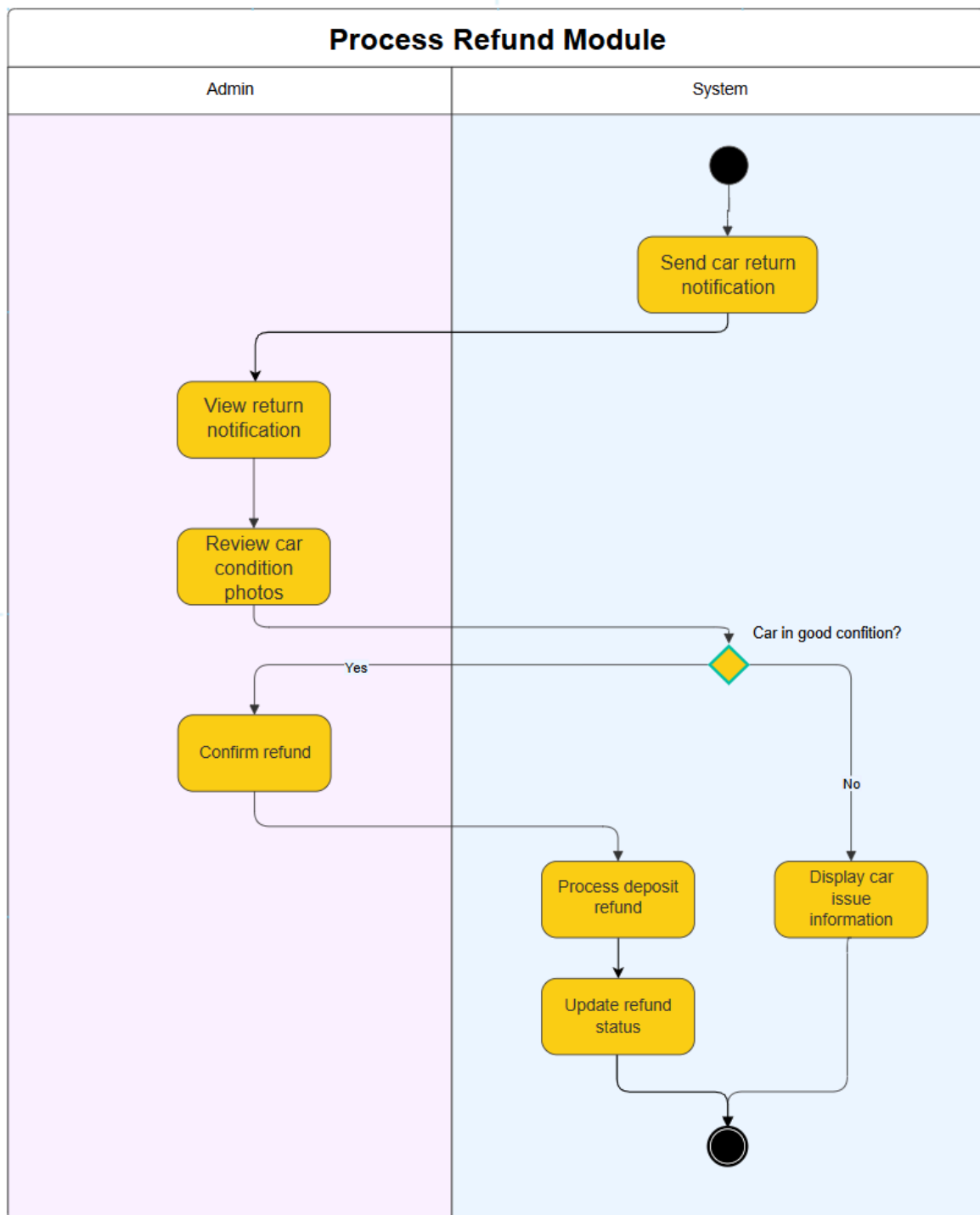


10. Manage Feedback

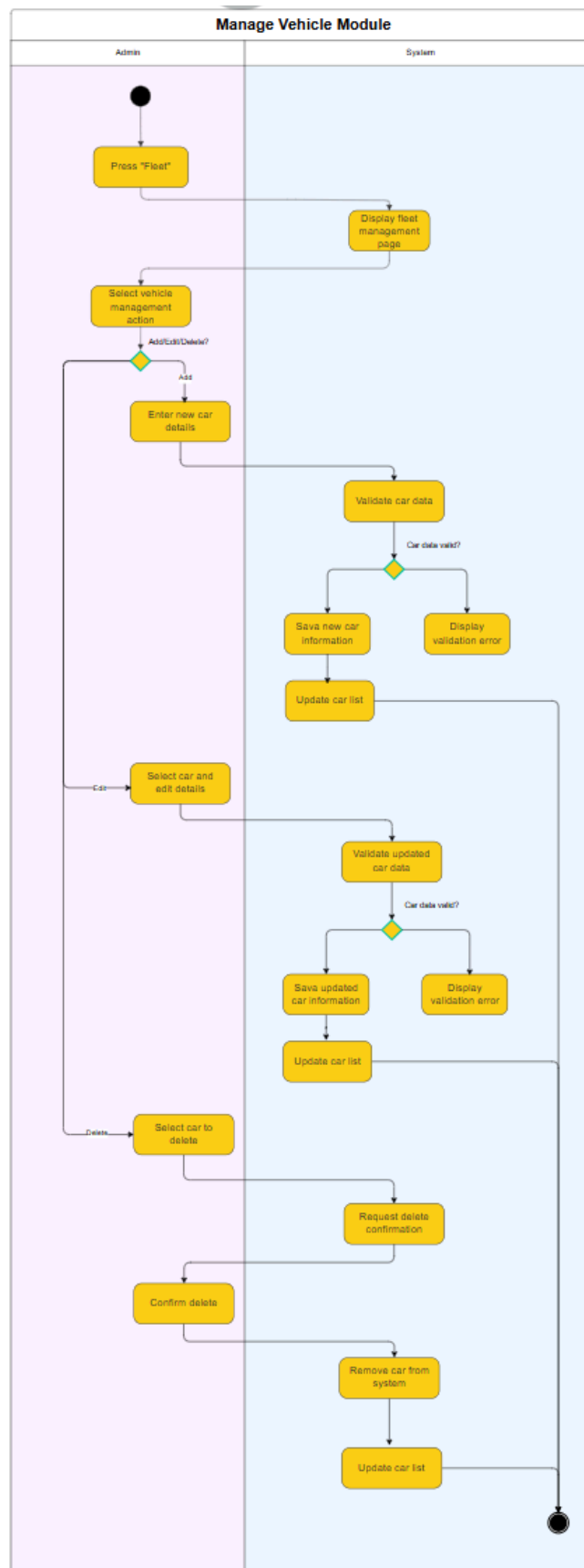




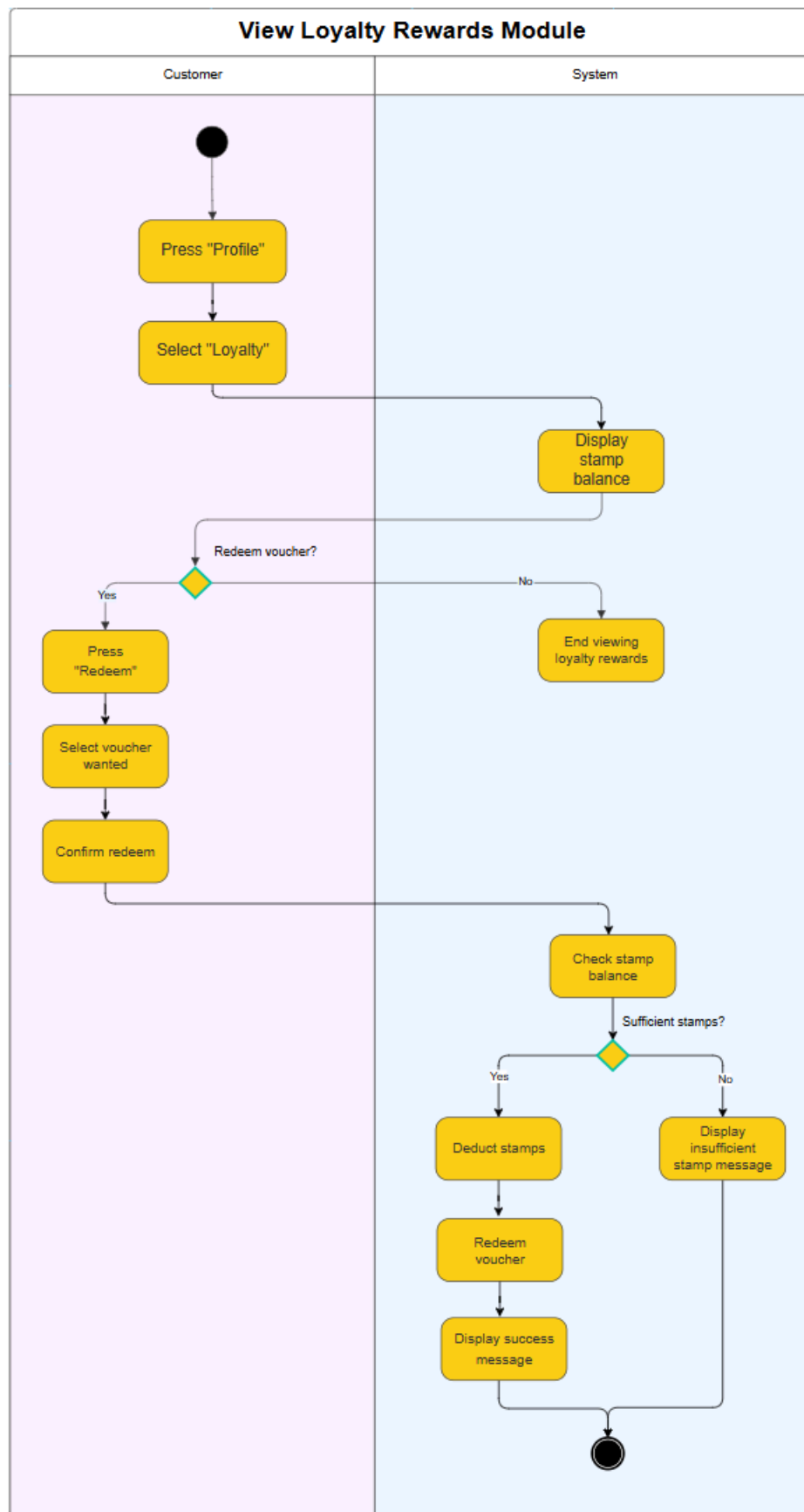
11. Process Refund



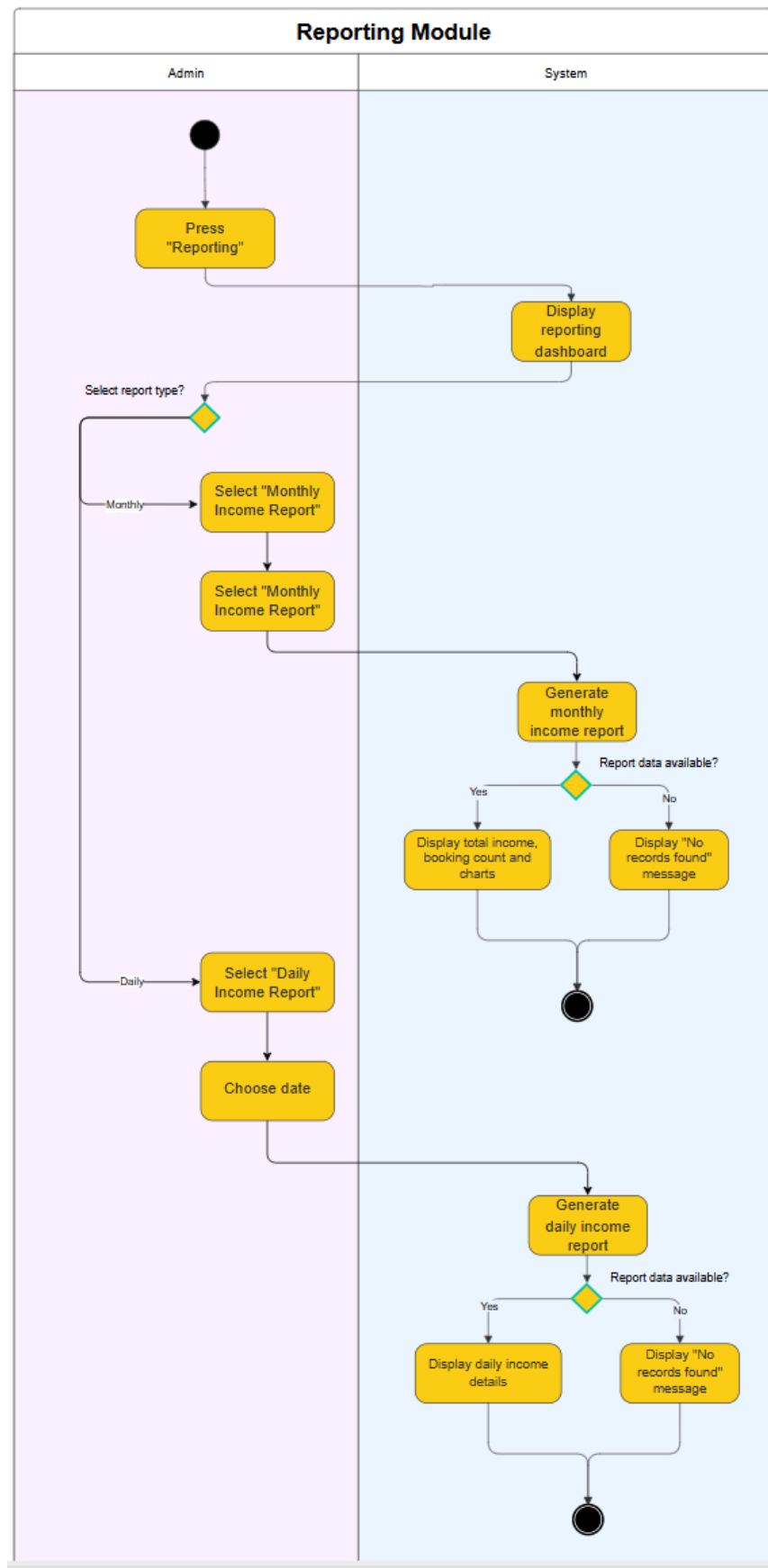
12. Manage Vehicle



13. View Loyalty Rewards



14. Reporting



7.3 Meeting Logs

MEETING MINUTES

Date/Time		14 January 2026 9:30pm	
Location		Google Meet	
Agenda		1. Review project requirements and scope for Hasta Travel Vehicle Rental Management System	
		2. Discuss and finalize system modules and their functionalities	
		3. Assign responsibilities for ERD development and documentation	
		4. Review Data Flow Diagrams (Context, Level 0, and Child Diagrams)	
		5. Establish timeline and milestones for deliverables	
Meeting MC		Angela Ngu Xin Yi	
Attendance			
Name		Time	Reason for Absence
Angela Ngu Xin Yi		2130	-
Gavenesh A/L Batumalai		2130	-
Tan Xin Tian		2130	-
Teoh Xin Yee		2130	-
Toh Shee Thong		2130	-
Minutes			
No.	Item Discussed	Ideas/Suggestions and Person Giving It	Person in Charge
1.	Project Scope and Objectives	Angela: Emphasized the need to improve Hasta Travel's manual booking system to eliminate WhatsApp-based inefficiencies and provide real-time vehicle	Angela, Gavenesh

		availability. Gavenesh: Suggested focusing on UTM-specific features like matric number verification and campus-based pickup locations. All members agreed on core features: automated booking, payment verification, loyalty rewards, and admin dashboard.	
2.	System Modules Definition	Xin Tian: Proposed six main modules - User Management, Fleet Management, Booking Management, Payment Management, Loyalty & Rewards, and Report & Analytics. Xin Yee: Recommended separating pickup and return processes as distinct entities in the database for better tracking. Shee Thong: Suggested including vehicle maintenance tracking and renewal reminders for road tax and insurance.	Xin Tian, Xin Yee and Shee Thong
3.	ERD and Database Design Assignment	Angela: Assigned ERD development tasks: - User Management Module: Xin Yee - Fleet Management Module: Xin Tian - Booking Management Module: Shee Thong - Payment Management Module: Gavenesh - Loyalty & Rewards Module: Angela - Integration and Enhanced ERD: All members to review together	Angela, Gavenesh

		Gavenesh: Suggested using specialization for User entity (Customer/Admin hierarchy). All agreed to maintain consistency in naming conventions and relationship cardinalities.	
4.	DFD Review and Validation	Xin Yee: Presented draft Context Diagram showing main external entities (Customer, Admin, Manager) and system boundary. Shee Thong: Reviewed Level 0 DFD covering 8 main processes from Login to Deposit Withdrawal. Xin Tian: Suggested adding data stores for each module to show data flow clearly. Group discussion: Refined child diagrams for complex processes like "Book the Car" and "Verify Booking" to show detailed sub-processes.	Xin Yee, Shee Thong, Xin Tian
5.	Business Rules and Constraints	All members brainstormed and documented business rules Xin Yee: User verification requirements - customers must upload IC, matric card, and driving license before booking. Xin Tian: Vehicle availability logic - maintenance status overrides booking availability. Shee Thong: Booking constraints -	All members

		24-hour cancellation policy, 30-minute grace period for late returns. Gavenesh: Payment rules - QR payment only, manual admin verification required. Angela: Loyalty program - 1 stamp per rental ≥ 9 hours, stamp redemption for vouchers.	
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